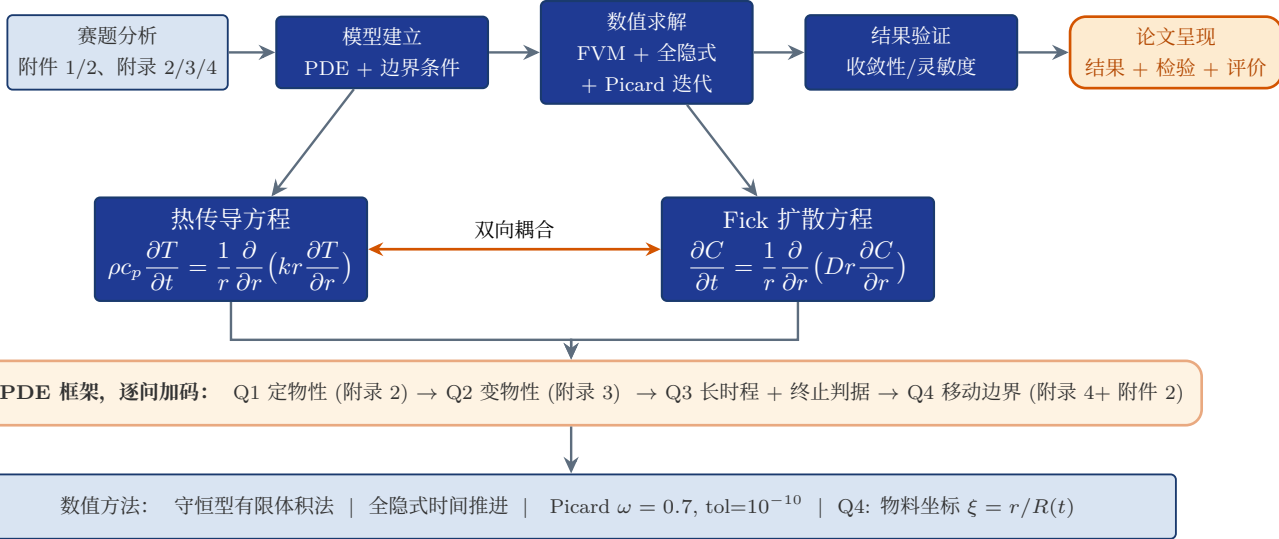
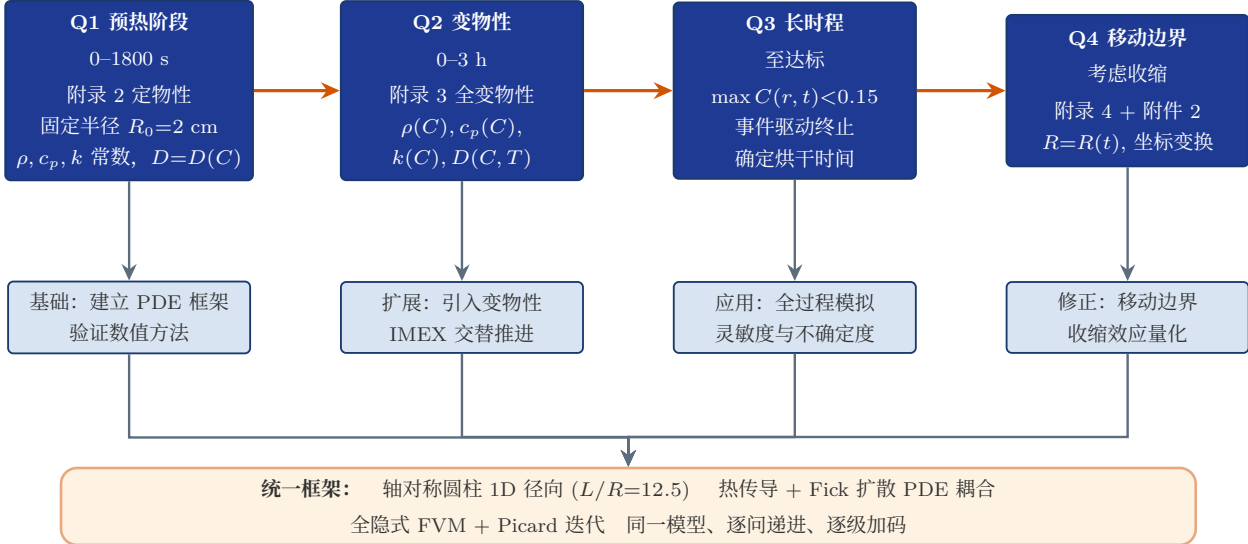
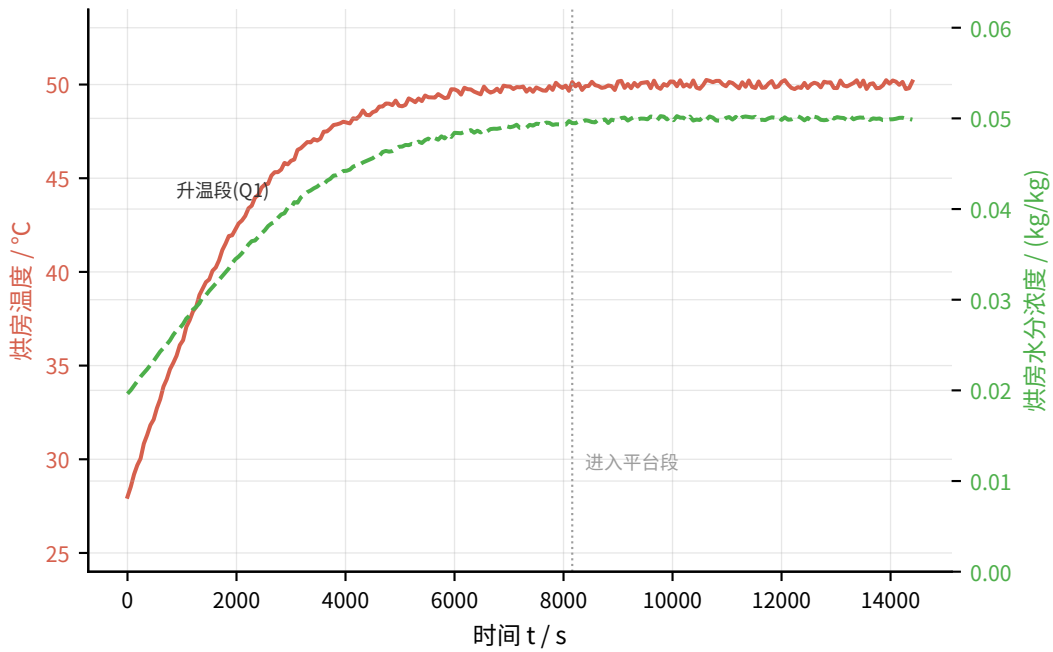


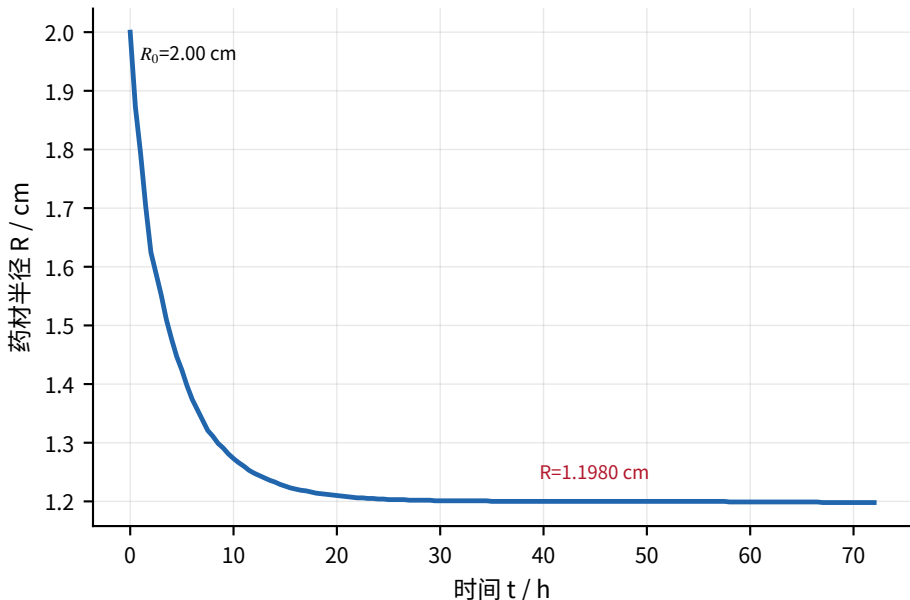




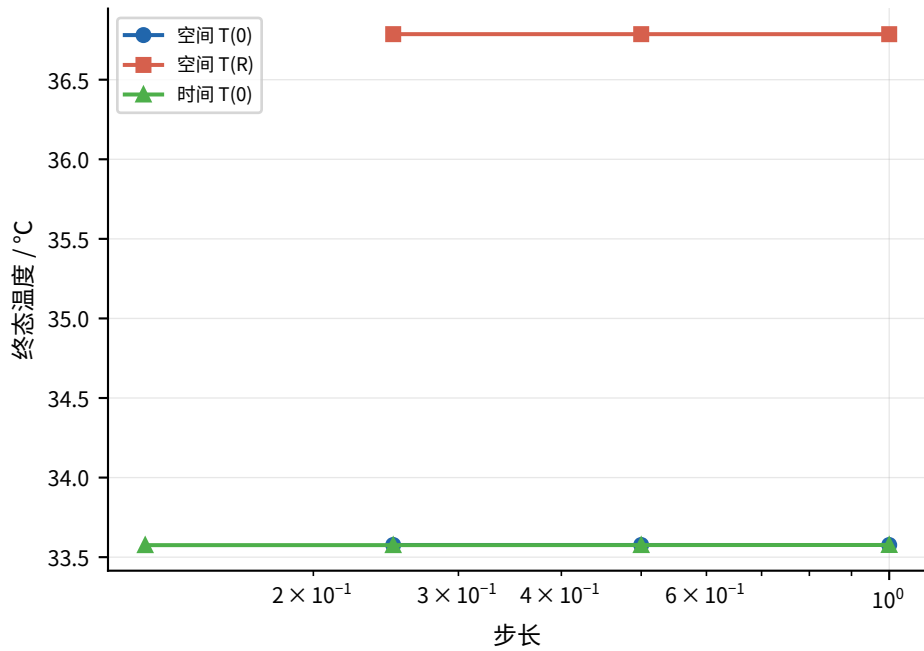
烘房环境条件时序 (附件1)



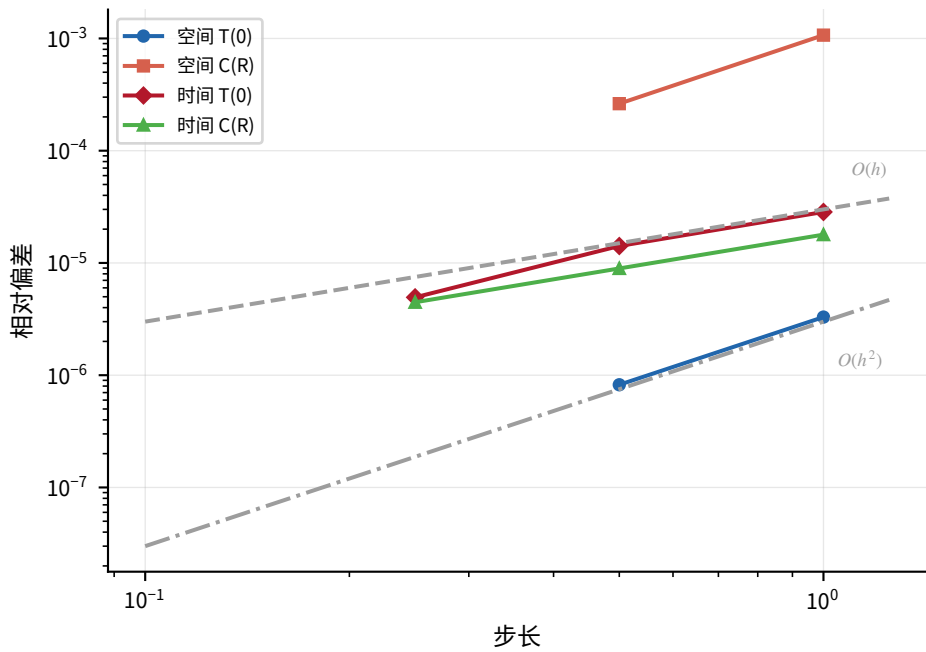
药材半径随干燥时间收缩（附件2）



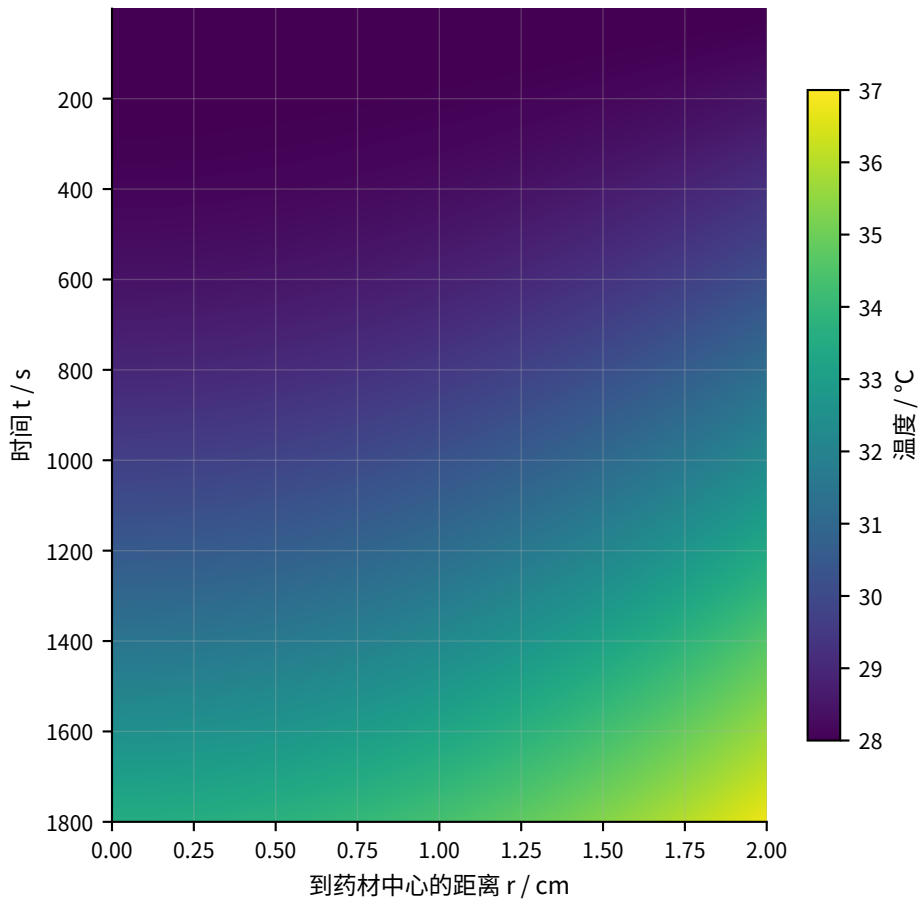
网格与时间收敛 — 终态值随步长



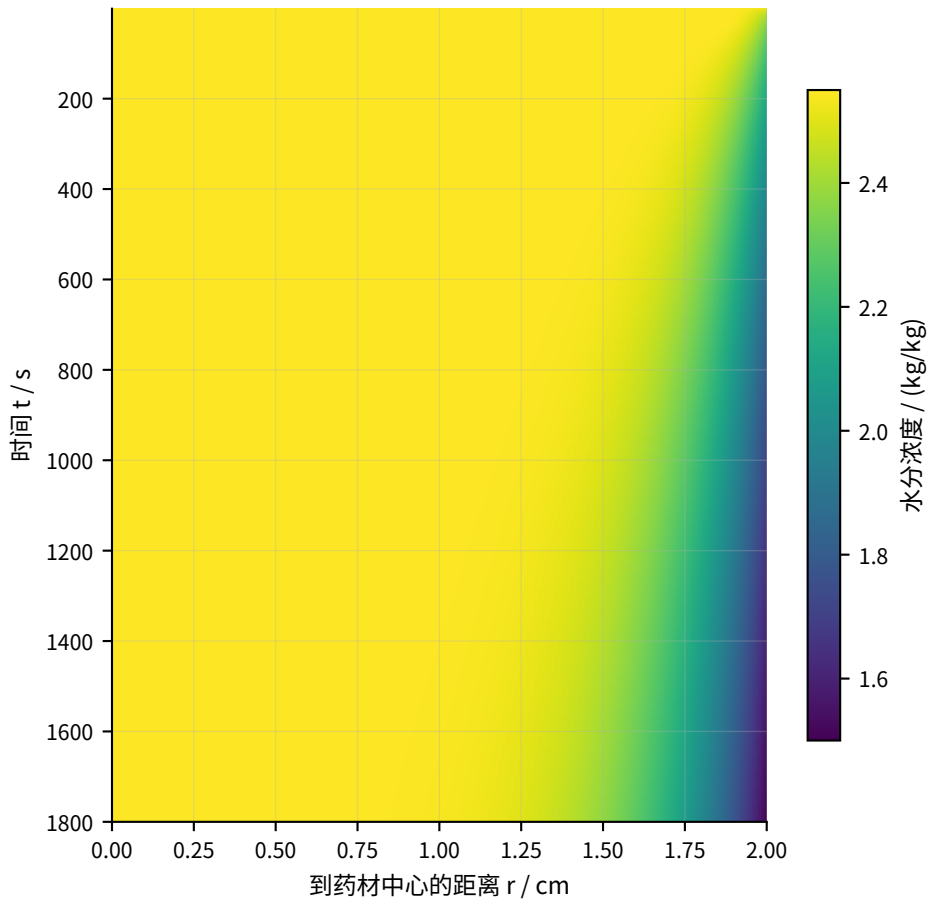
网格与时间收敛 — 收敛阶



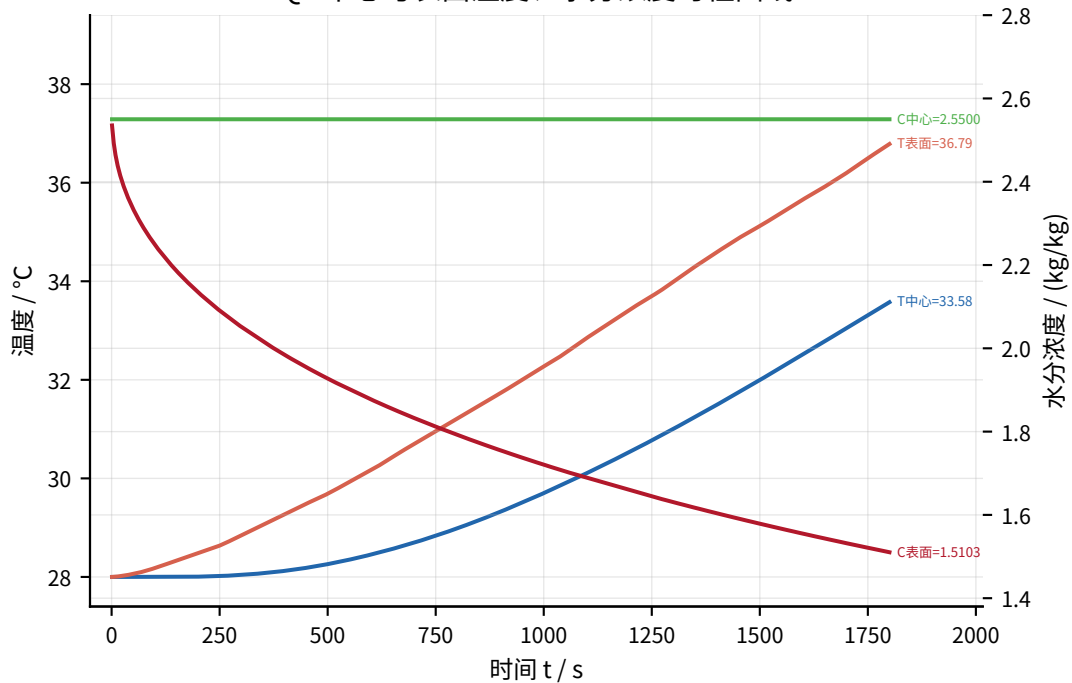
Q1 温度场 $T(r,t)$ 时空分布



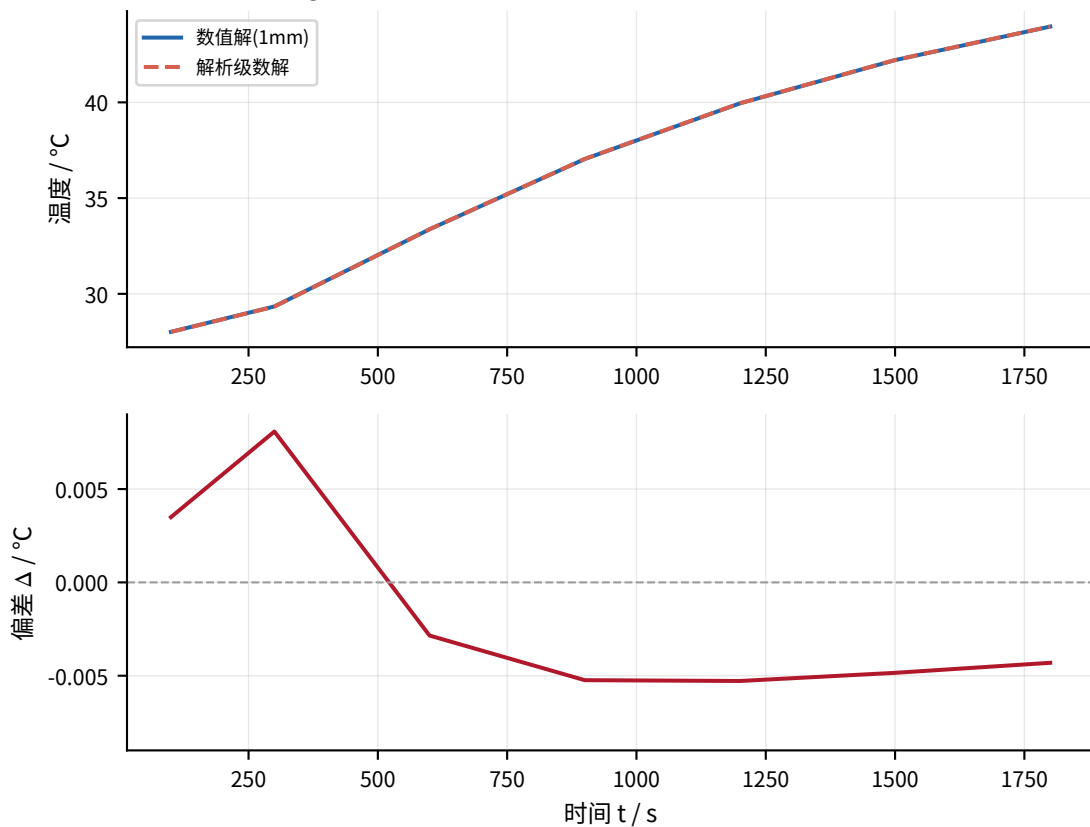
Q1 水分浓度场 $C(r,t)$ 时空分布



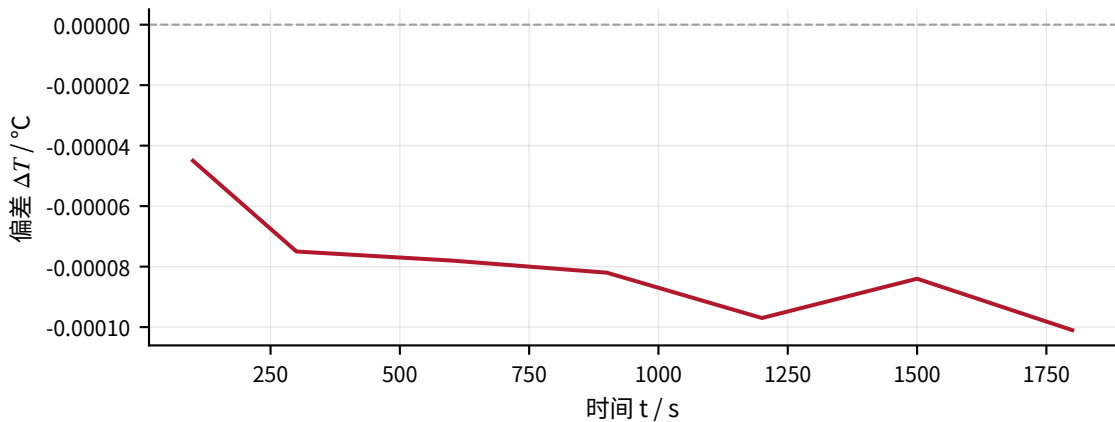
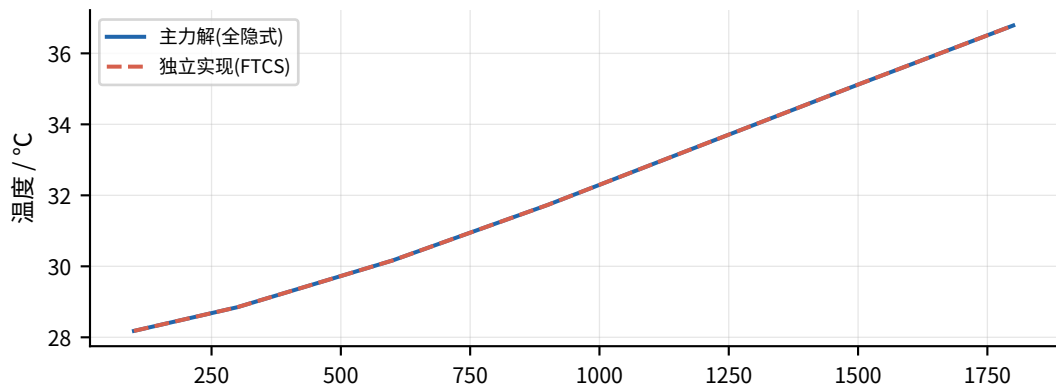
Q1 中心与表面温度、水分浓度时程曲线



Q1 温度解析对拍 (解析级数解 vs 数值解, $r=0$)

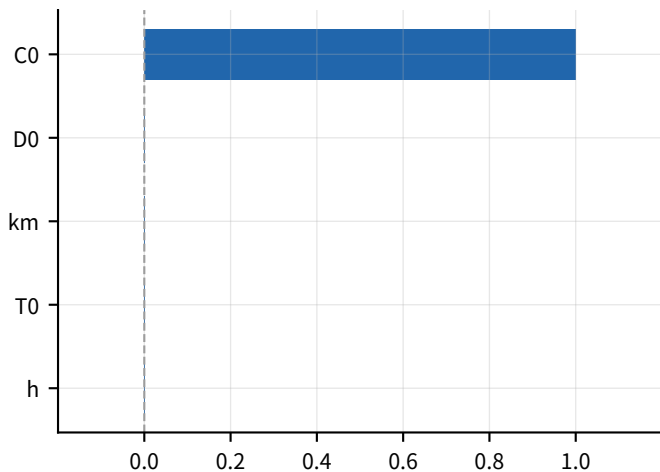


Q1 独立实现互验（表面温度， $t \leq 3h$ ）

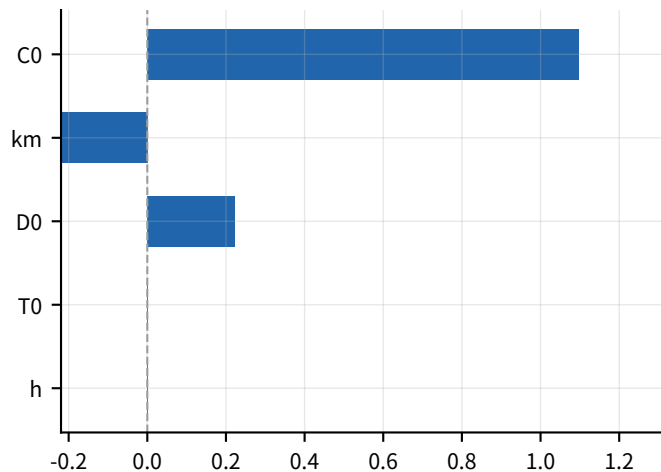


Q1 参数灵敏度龙卷风图 (OAT±20%)

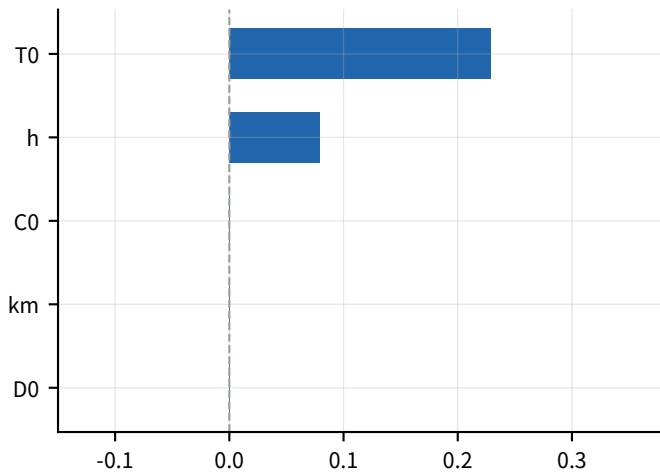
水分浓度 中心



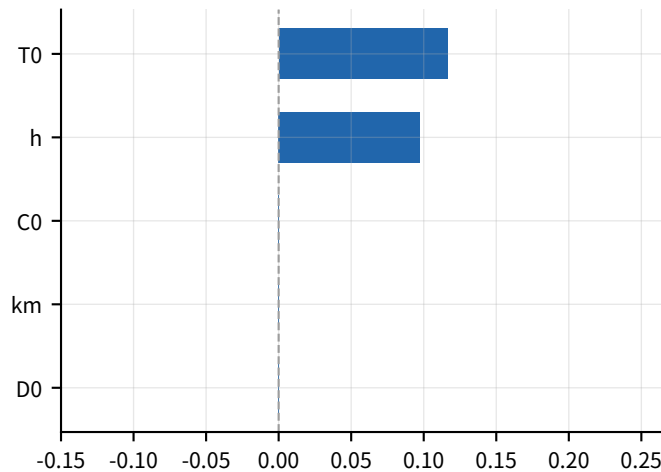
水分浓度 表面



温度 中心



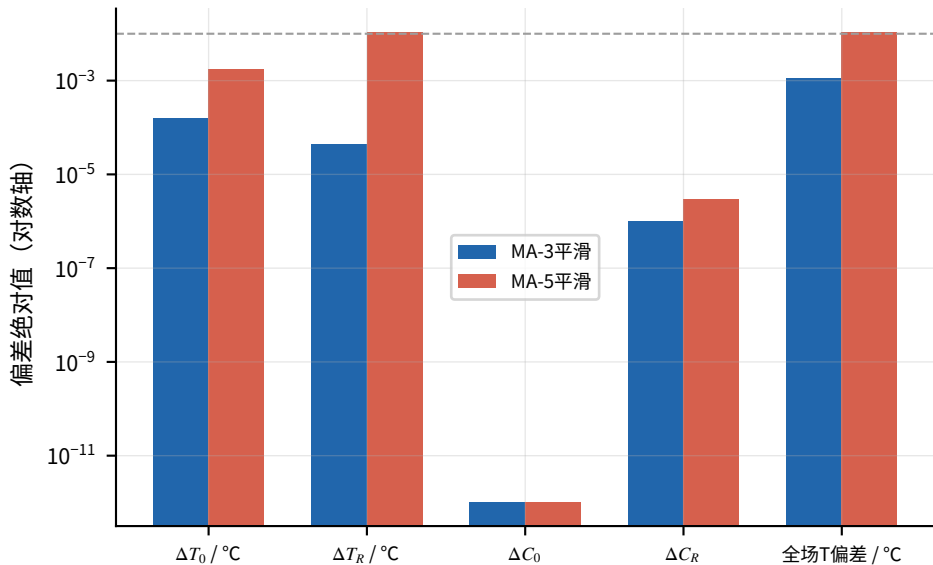
温度 表面



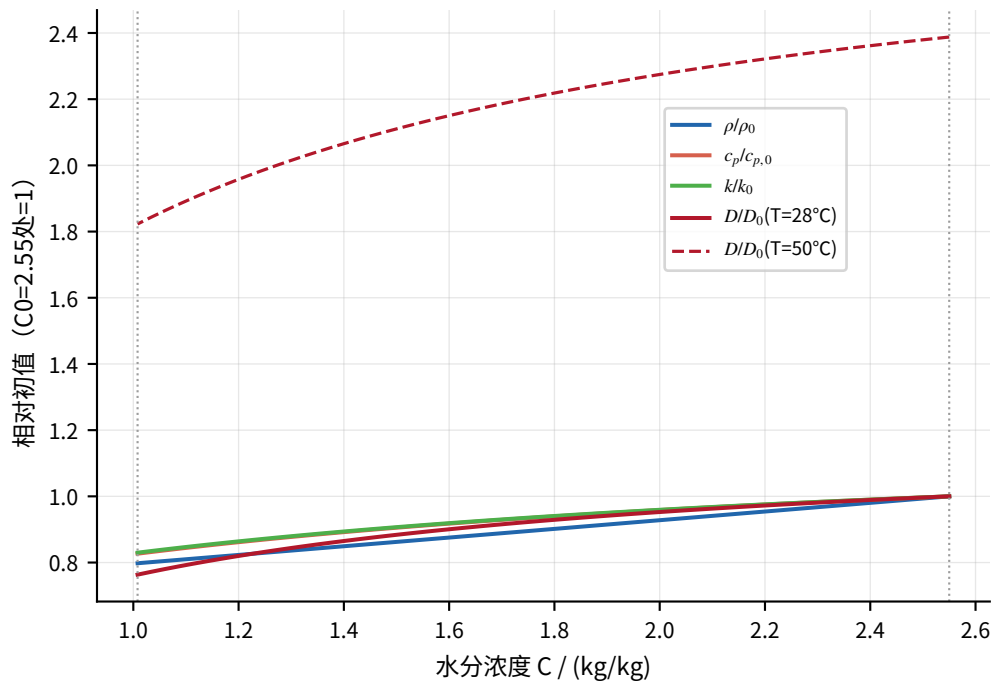
归一化灵敏度 S

归一化灵敏度 S

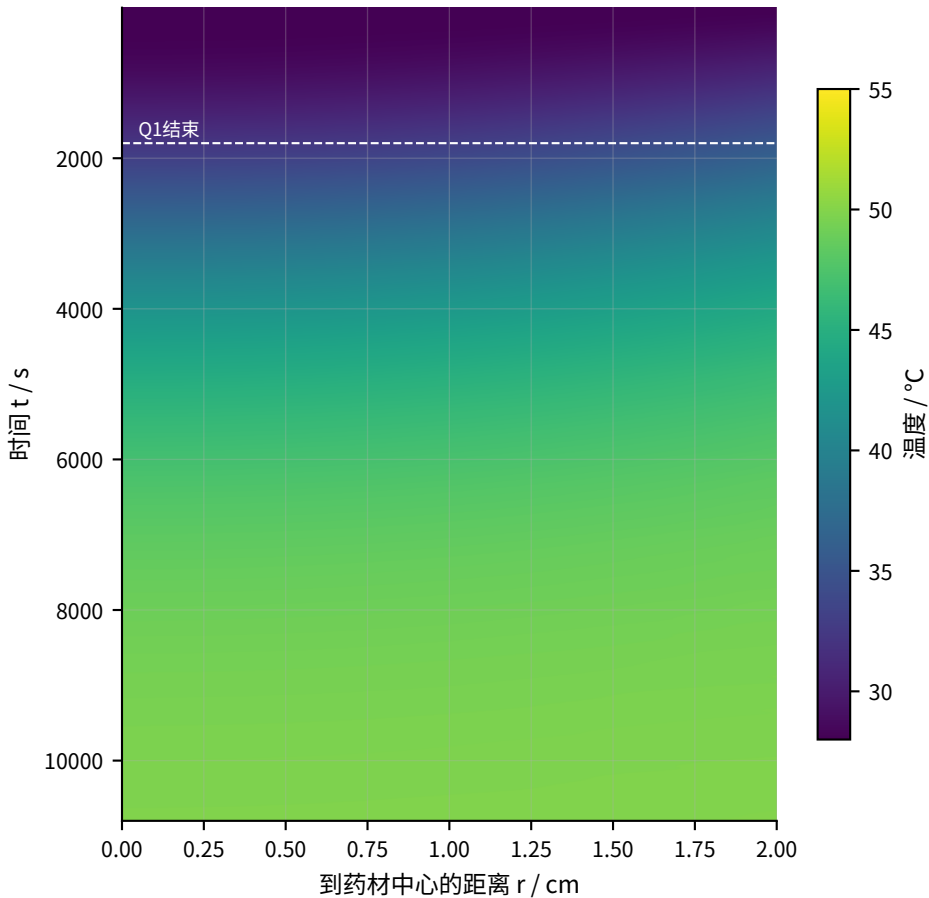
Q1 边界平滑 vs 不平滑对比



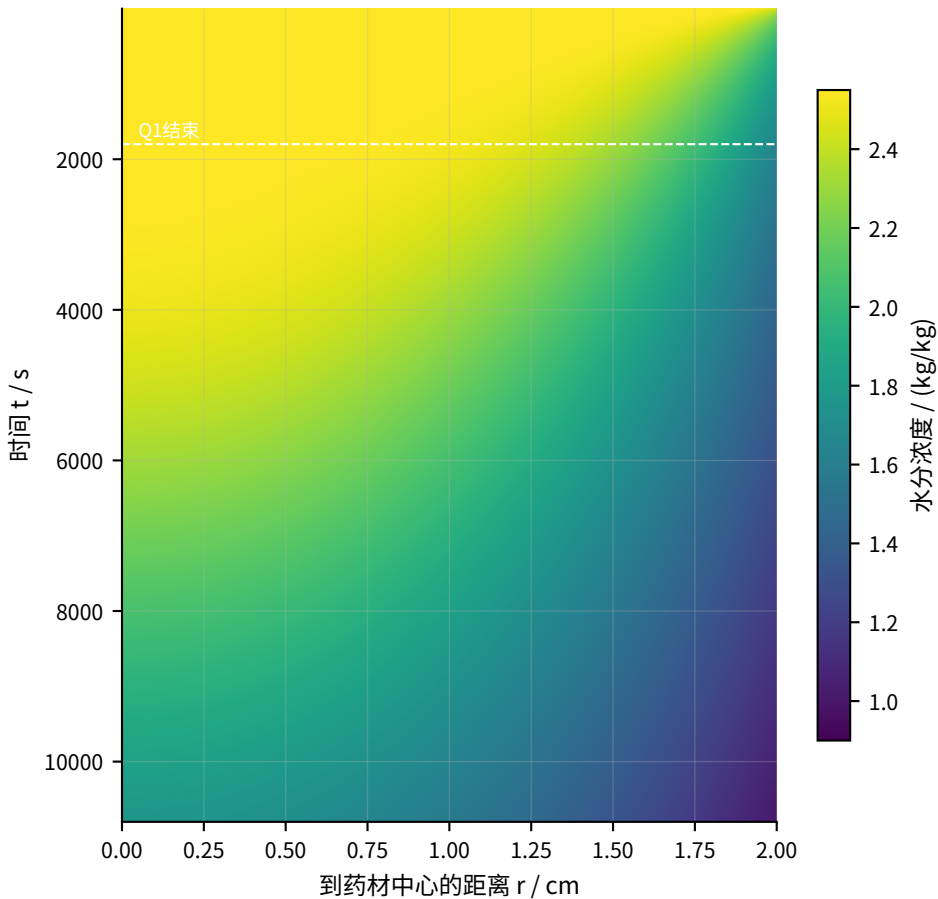
Q2 物性参数随水分浓度的演化



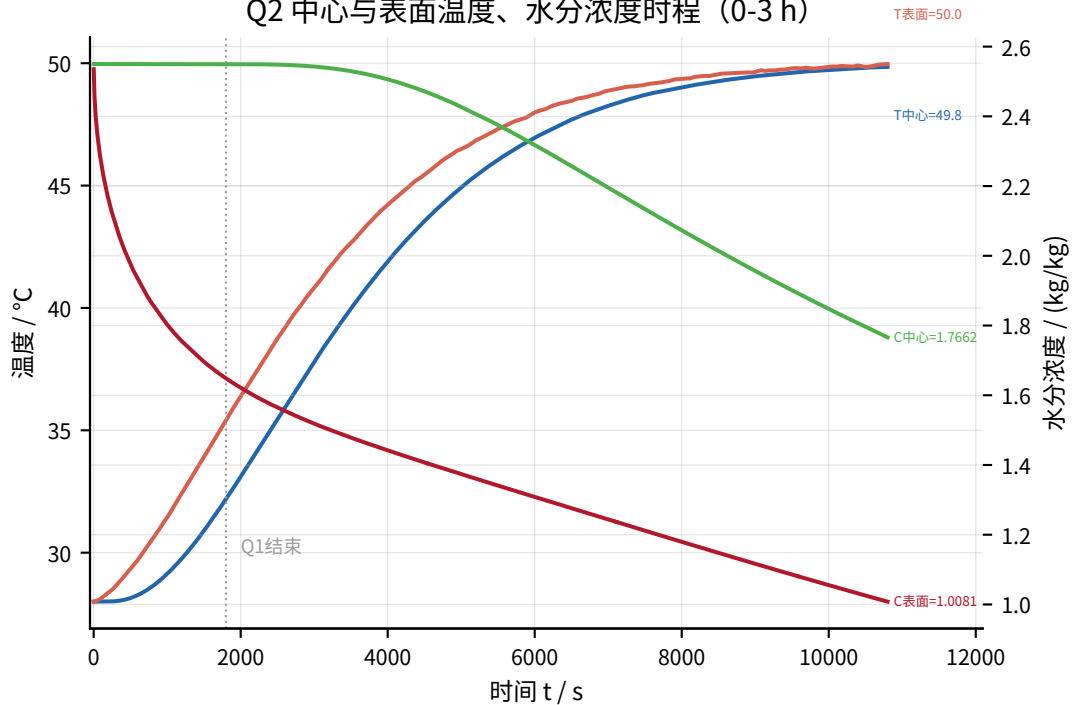
Q2 温度场 $T(r,t)$ 时空分布 (0-3 h)



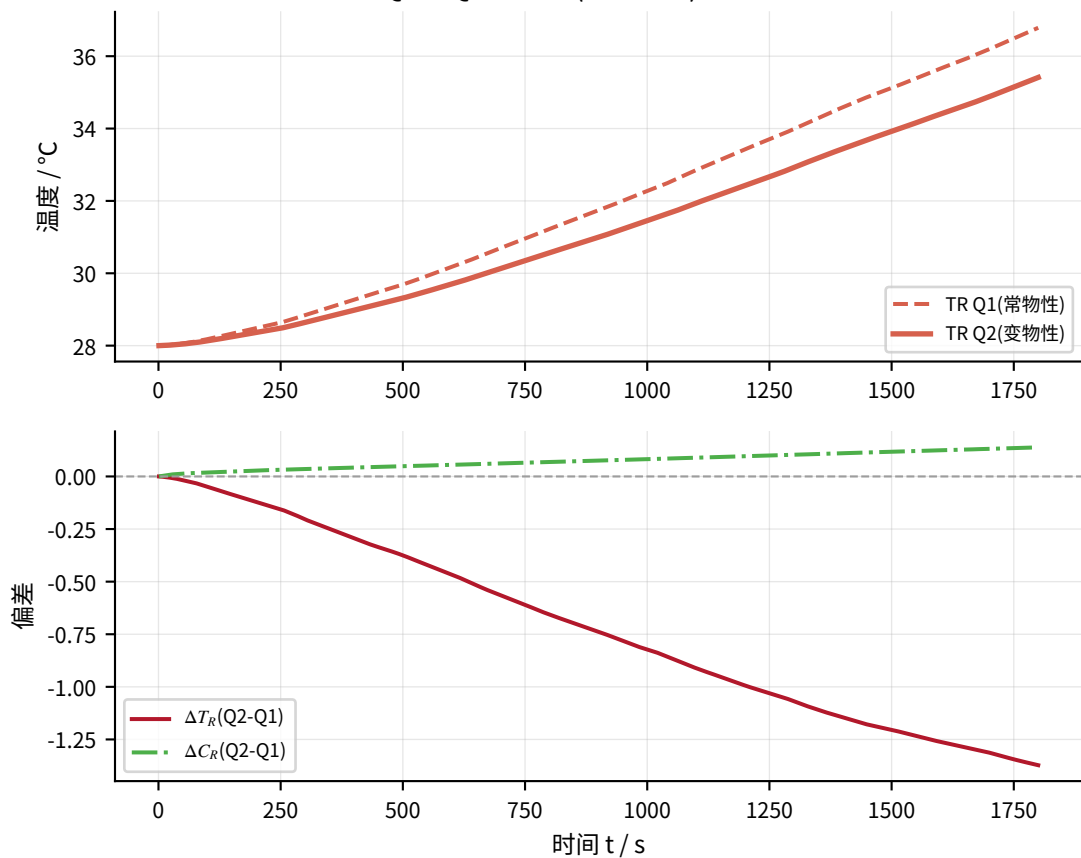
Q2 水分浓度场 $C(r,t)$ 时空分布 (0-3 h)



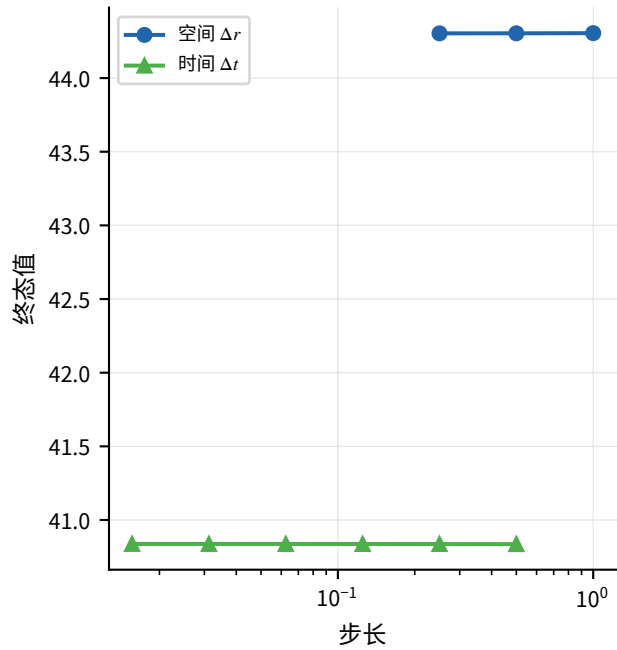
Q2 中心与表面温度、水分浓度时程 (0-3 h)



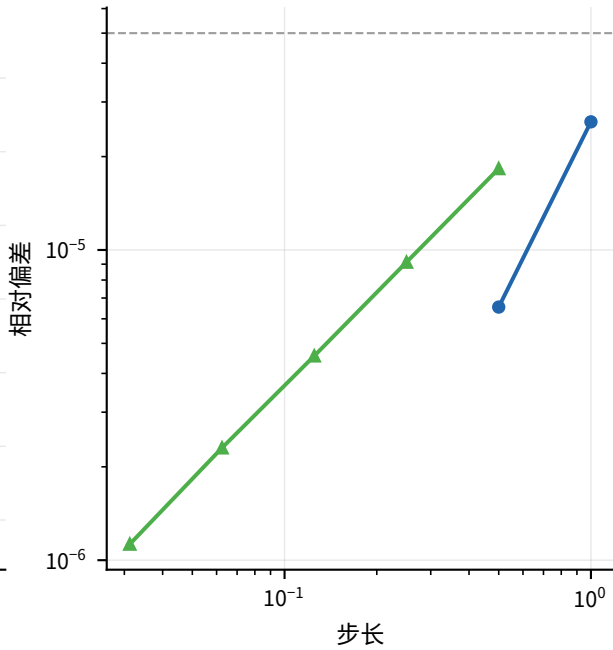
Q1与Q2重叠段(0-1800s)差异归因



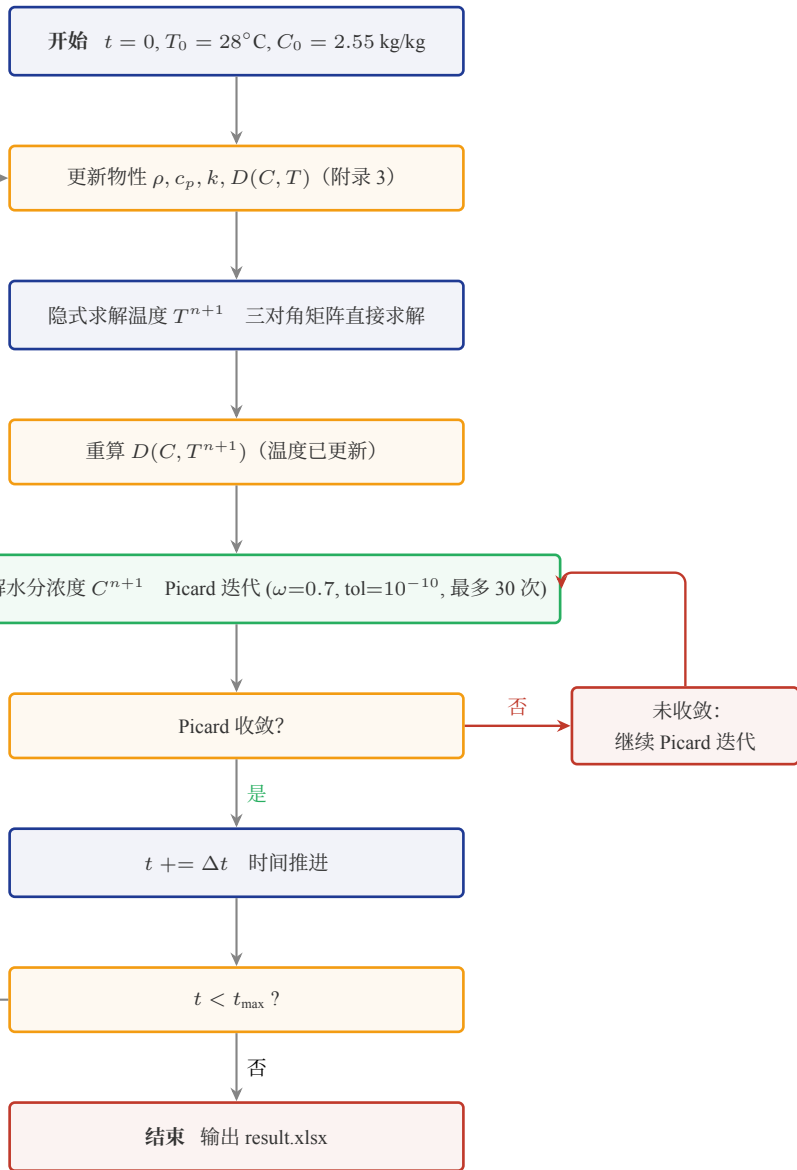
(a) 收敛趋势



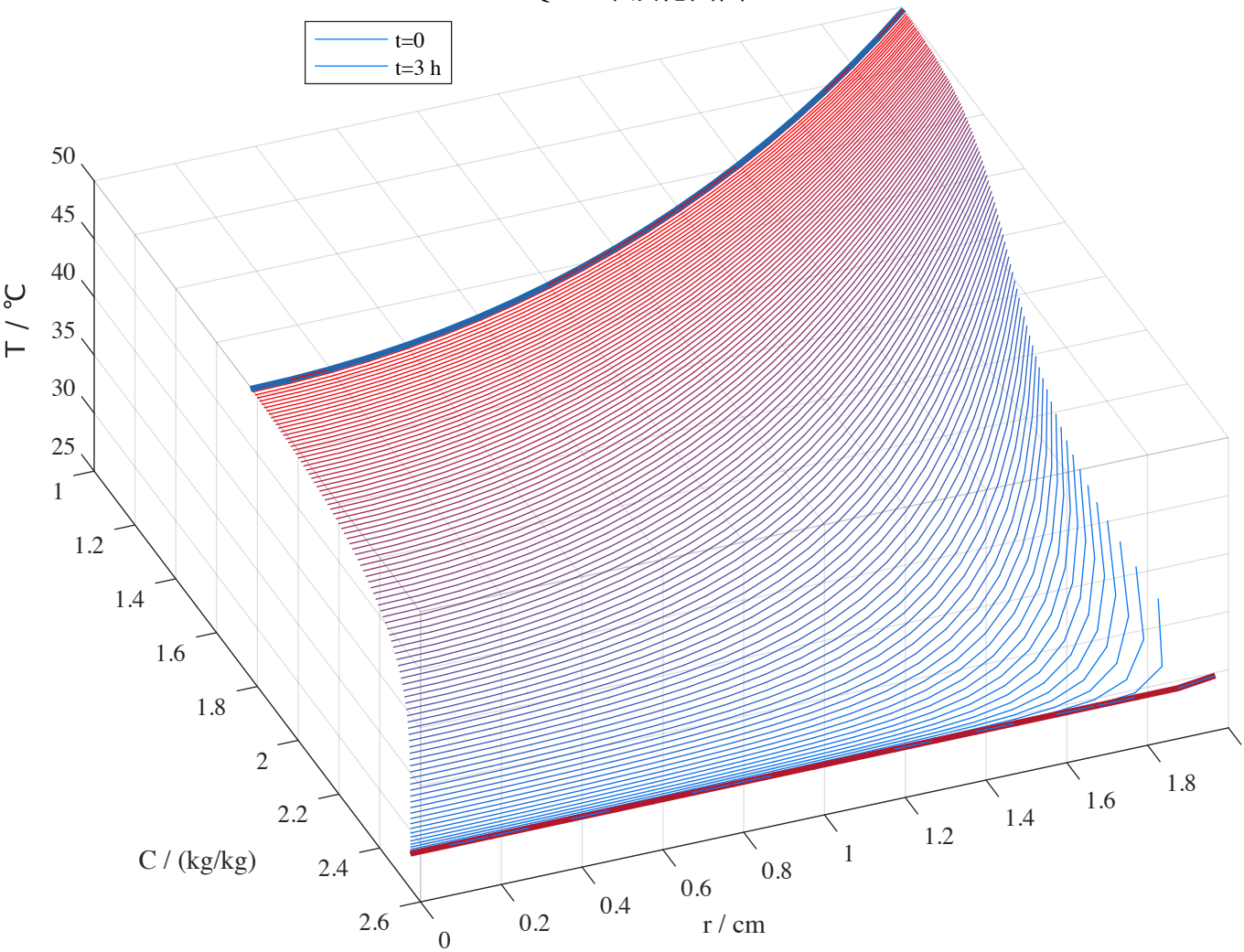
(b) 收敛阶



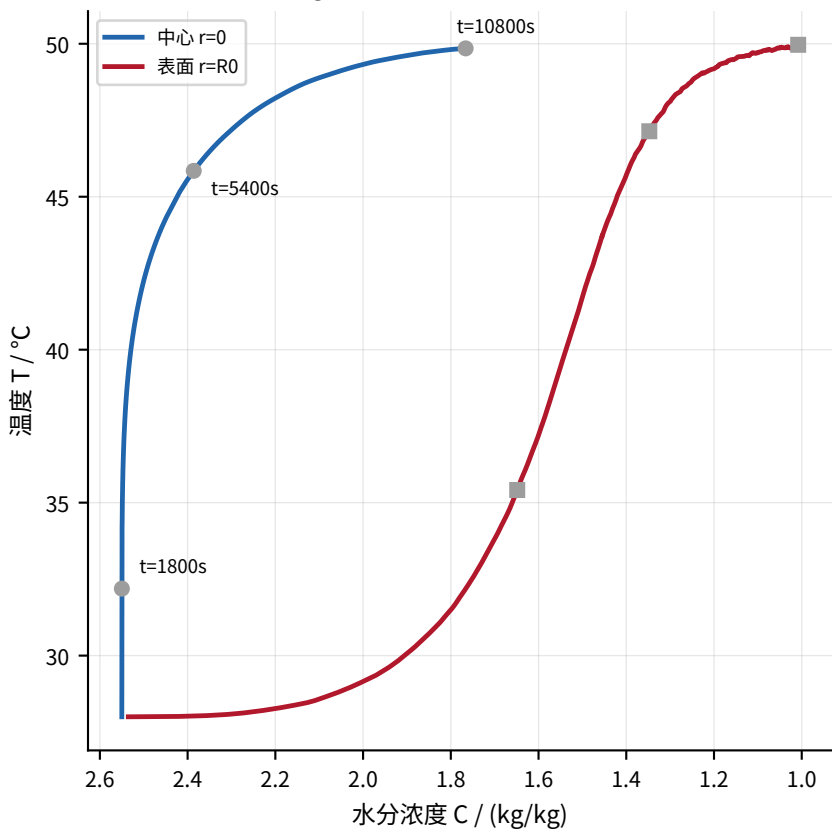
是



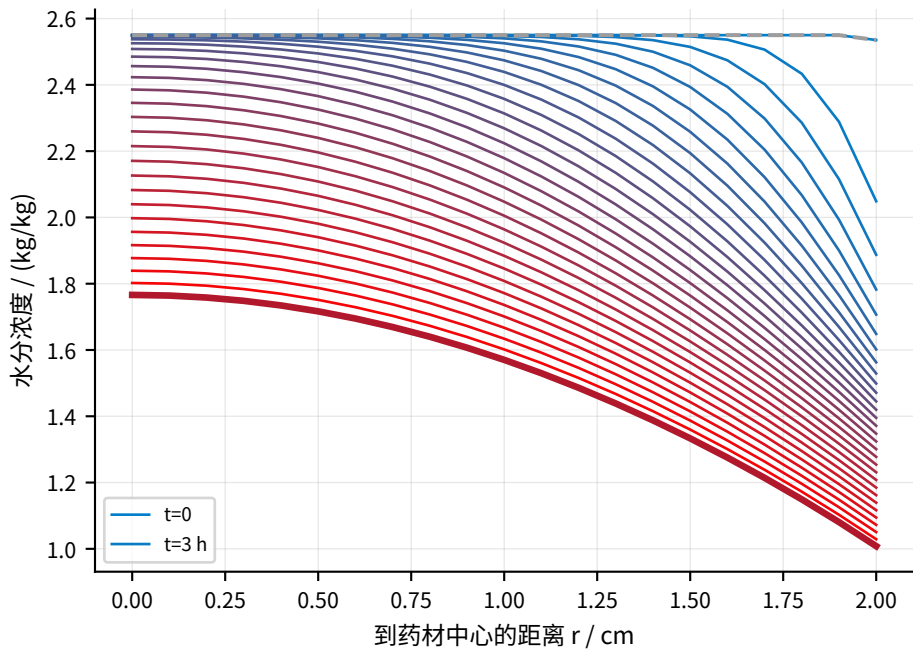
Q2 三维演化曲面 T-C-t



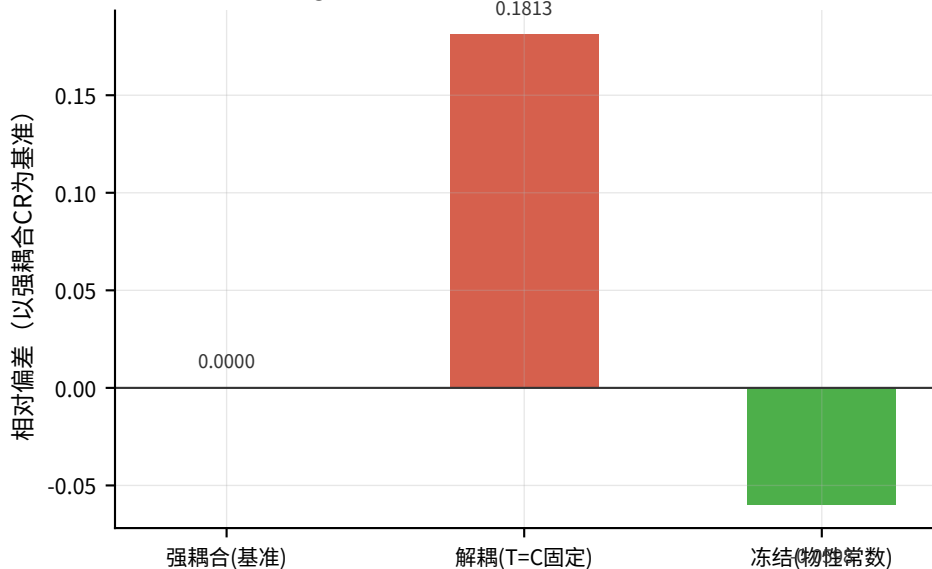
Q2 温度-水分浓度相轨迹



Q2 全断面失水演化 (浅→深=时间推进)

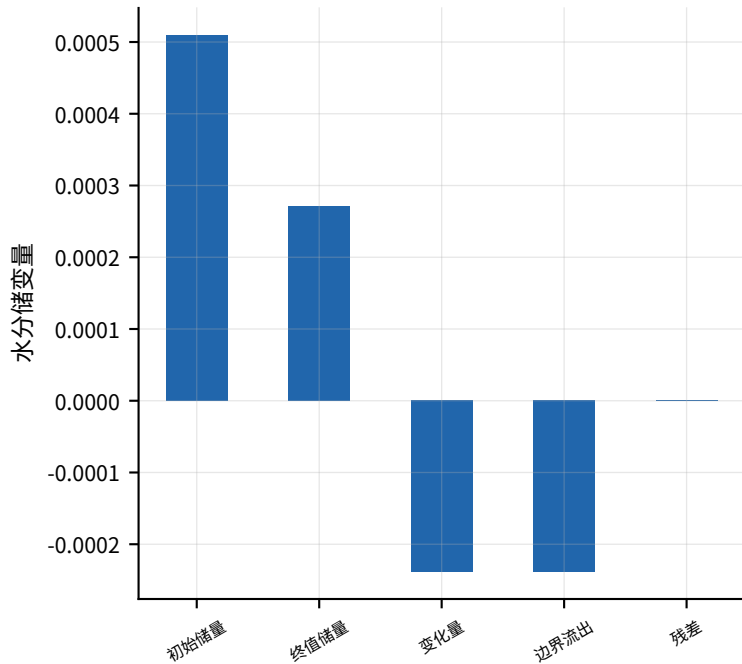


Q2 耦合强度分解 (3 h末CR偏差)

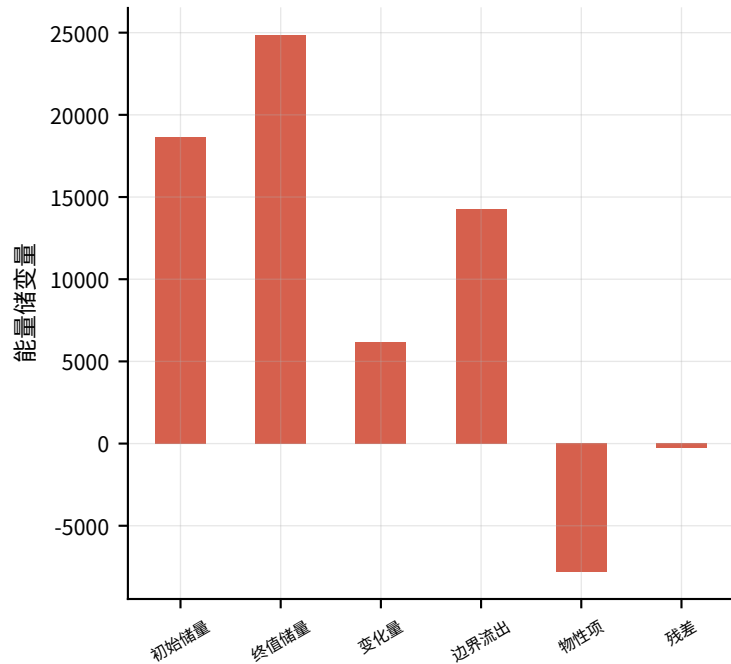


Q2 守恒性总账

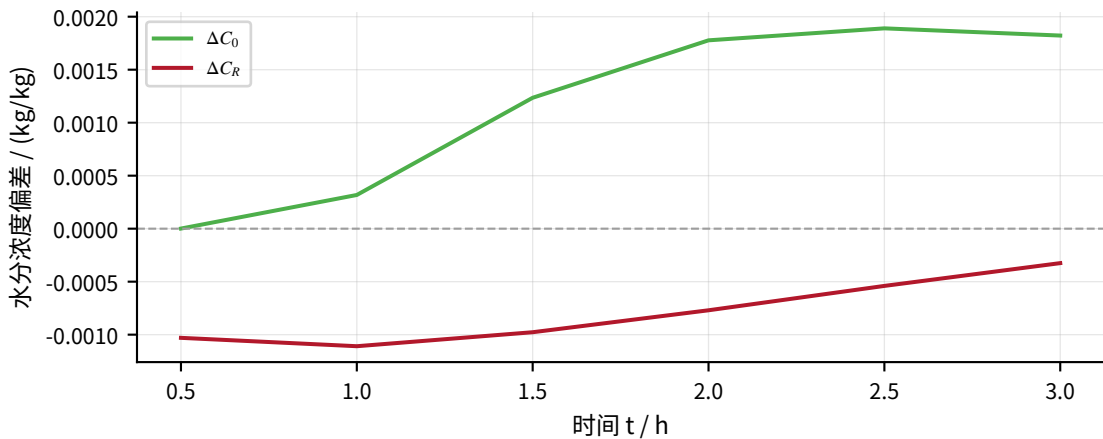
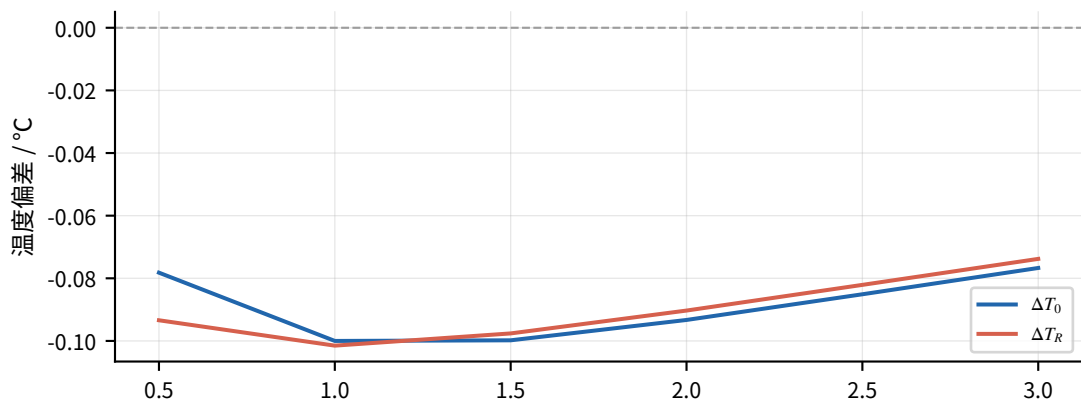
质量守恒



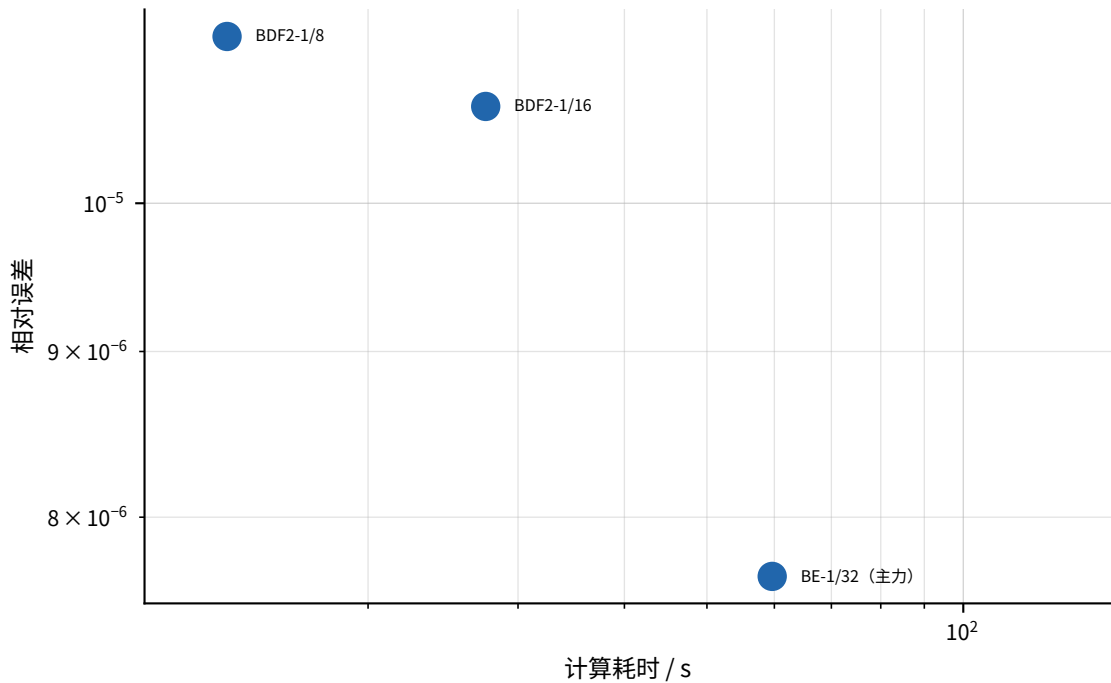
能量守恒



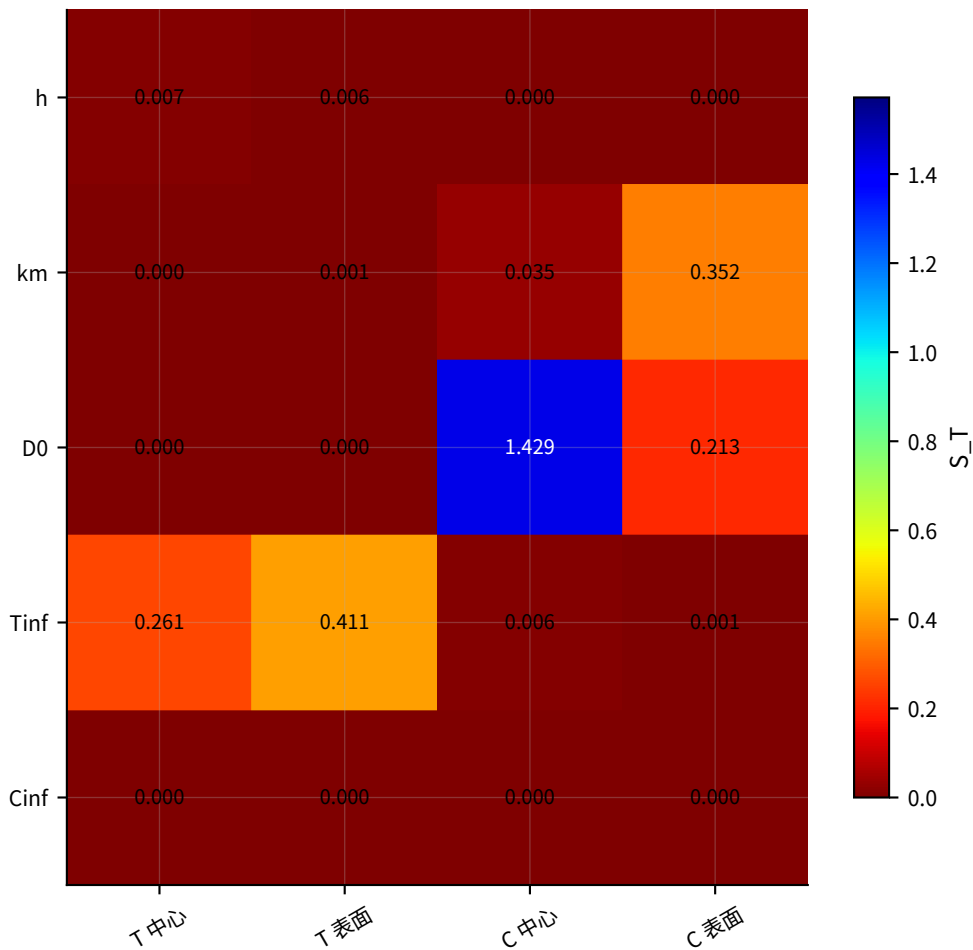
潜热效应对照（计入潜热 - 忽略潜热）



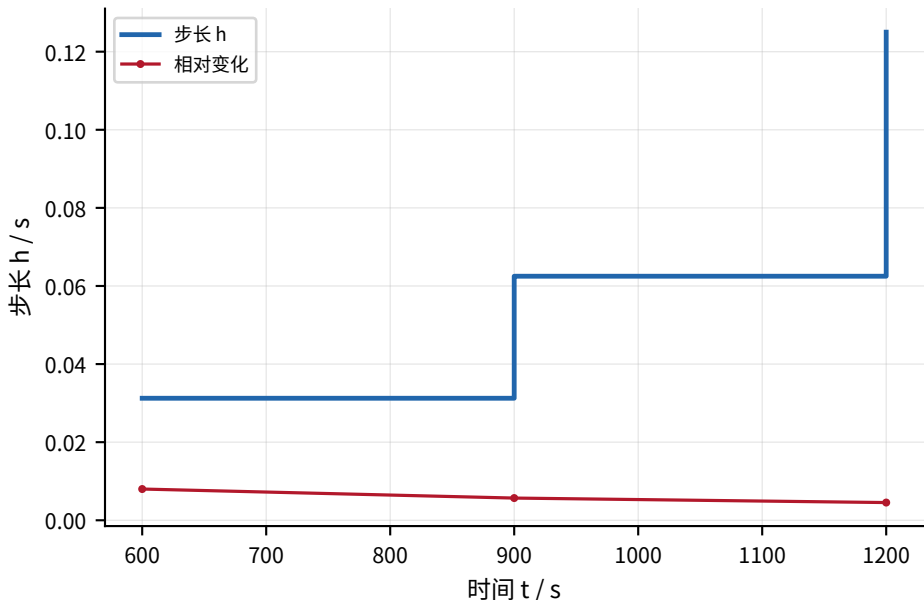
Q2 精度-效率帕累托图



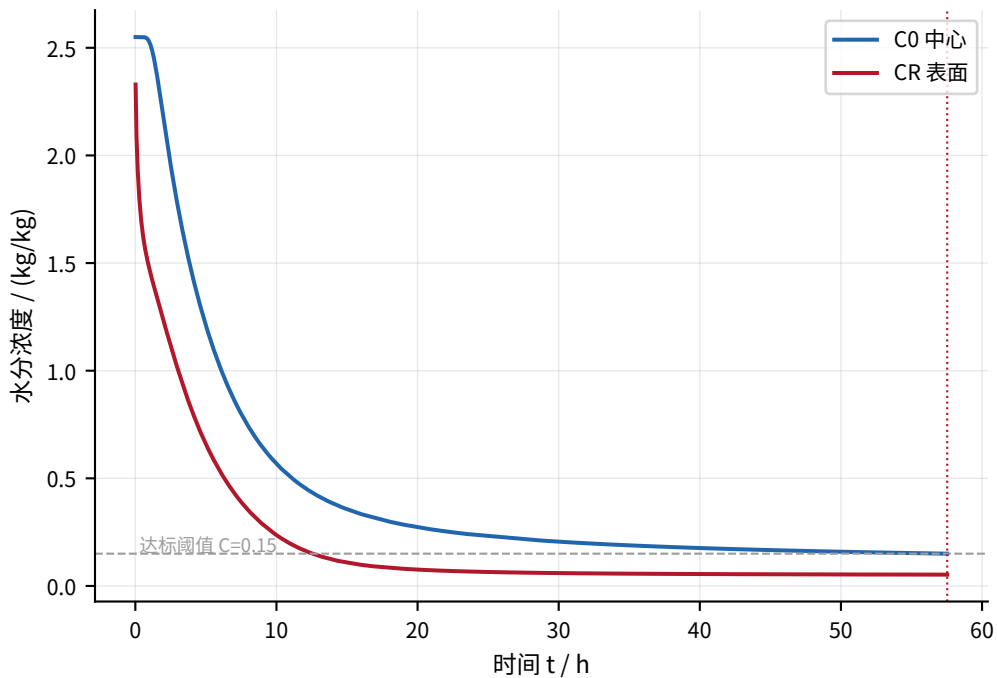
Q2 全局灵敏度 Sobol ST 热力图



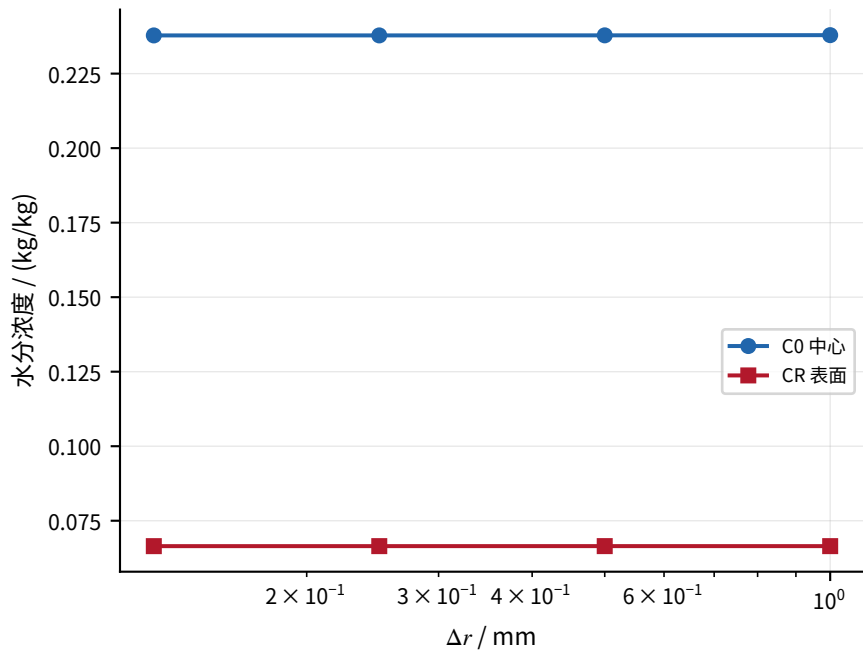
Q2 自适应步长轨迹



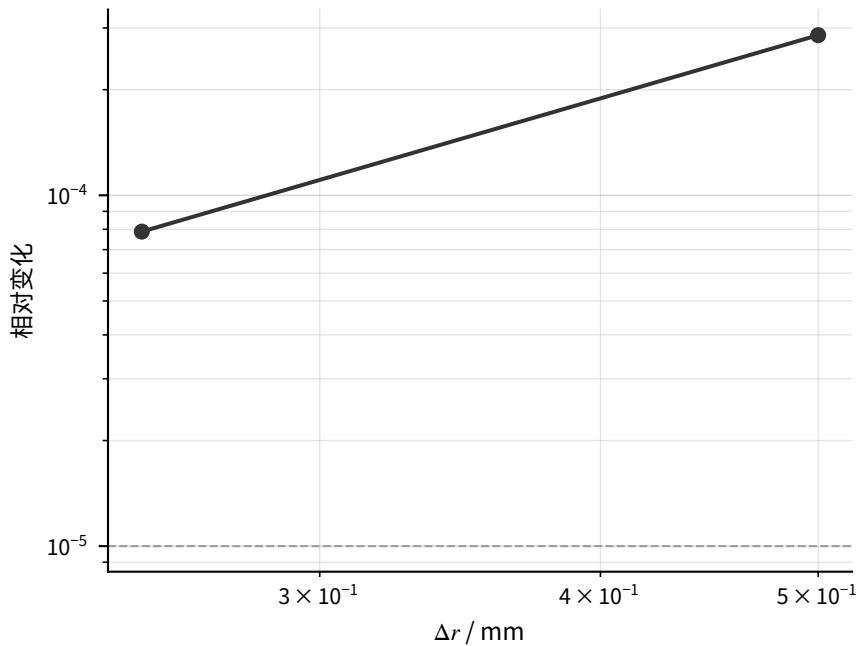
Q3 中心与表面水分浓度长时程演化



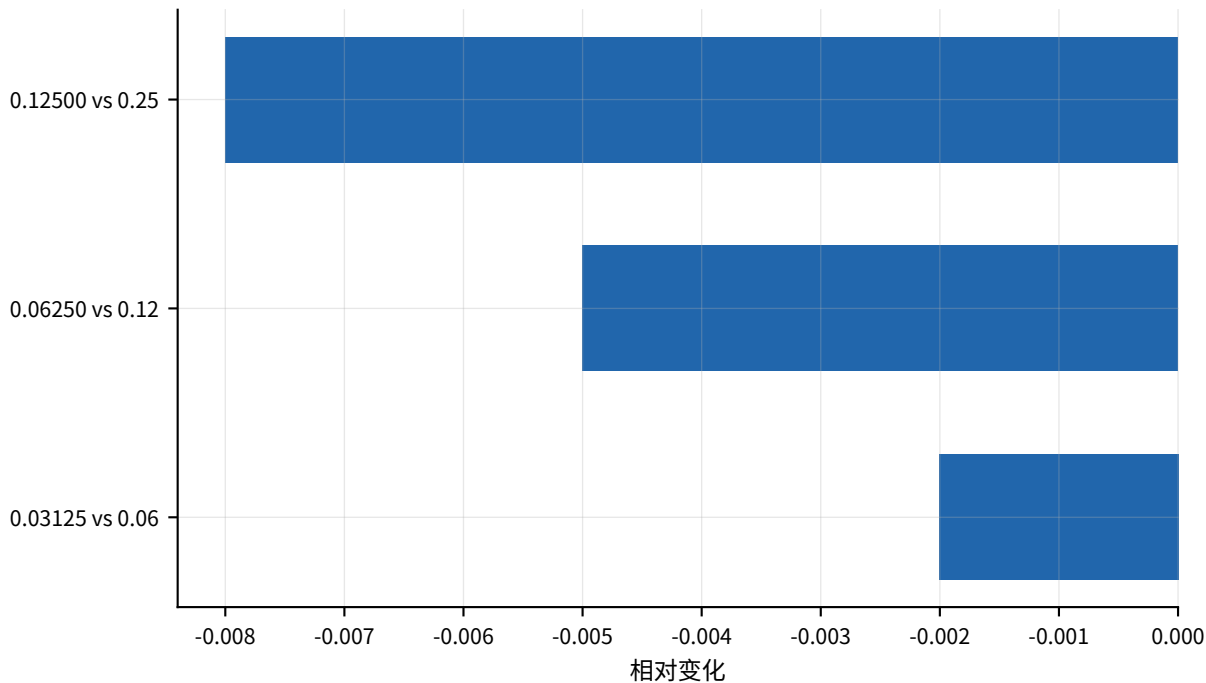
Q3 收敛性 — 终态值随步长



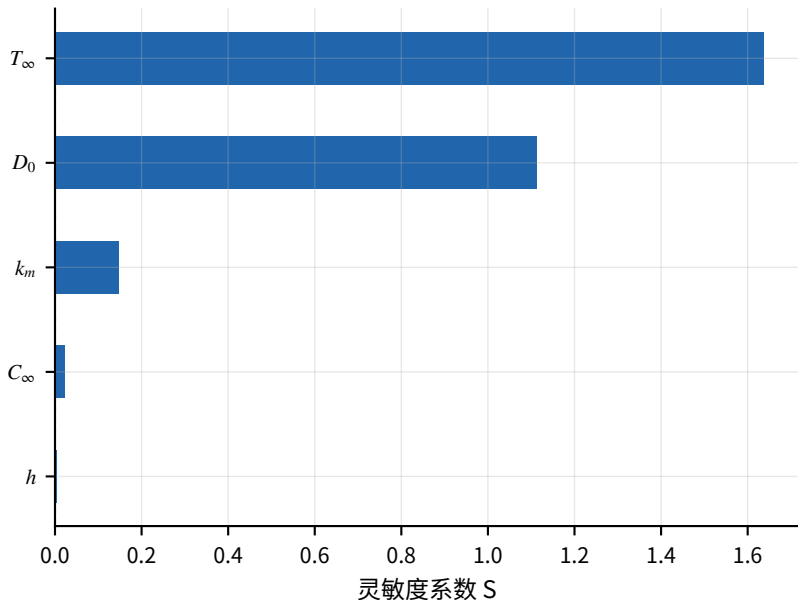
Q3 收敛性 — 收敛阶



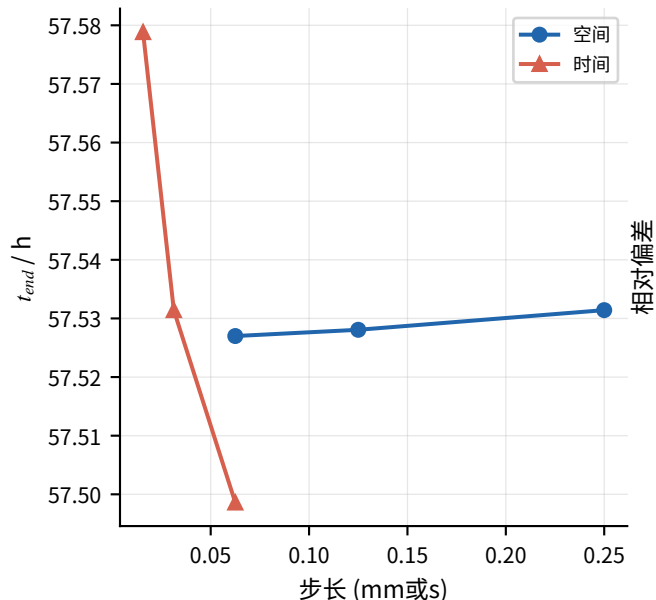
Q3 收敛性 — 各位置收敛情况



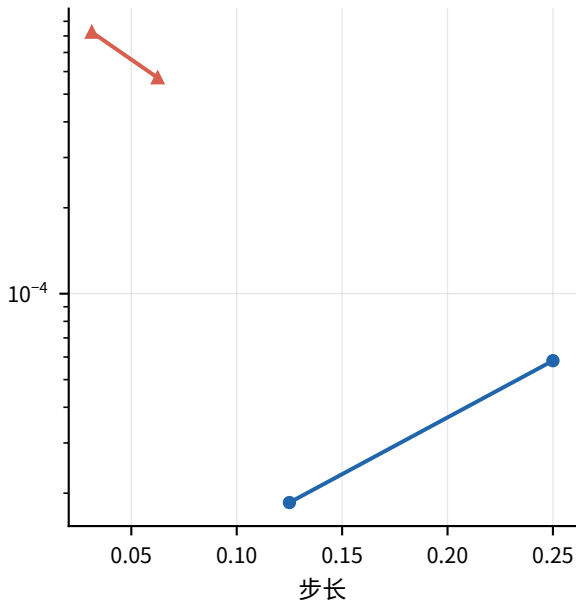
Q3 参数灵敏度



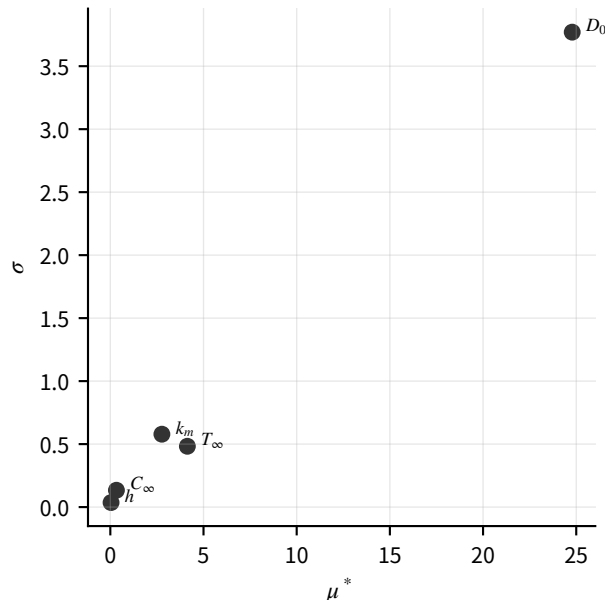
(a) 终态值



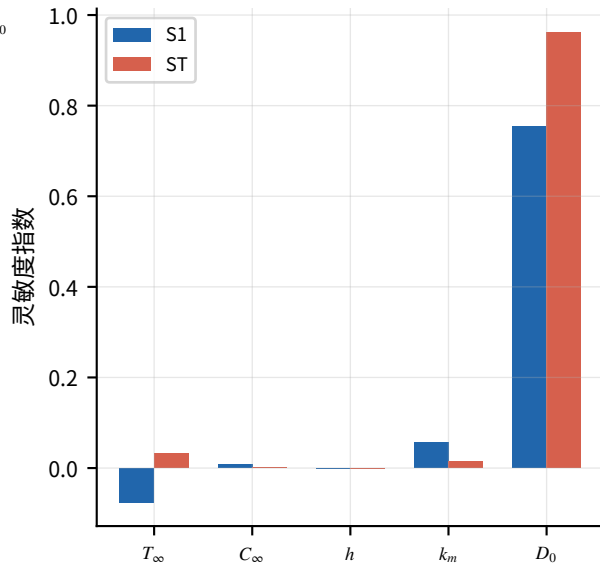
(b) 收敛趋势



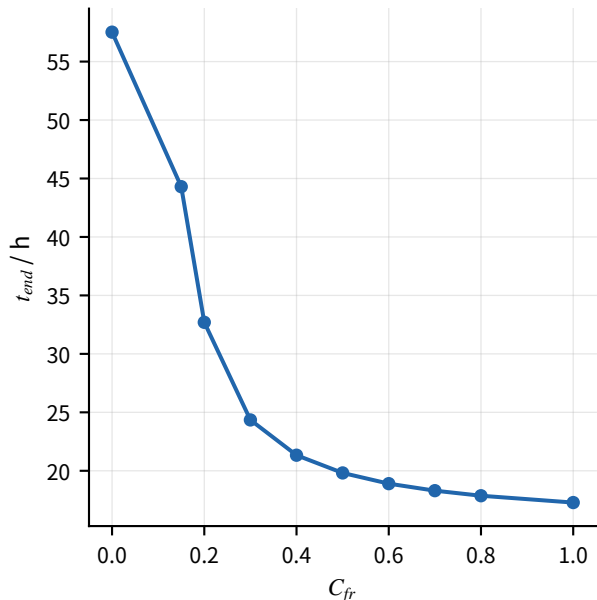
Morris $\mu^* - \sigma$



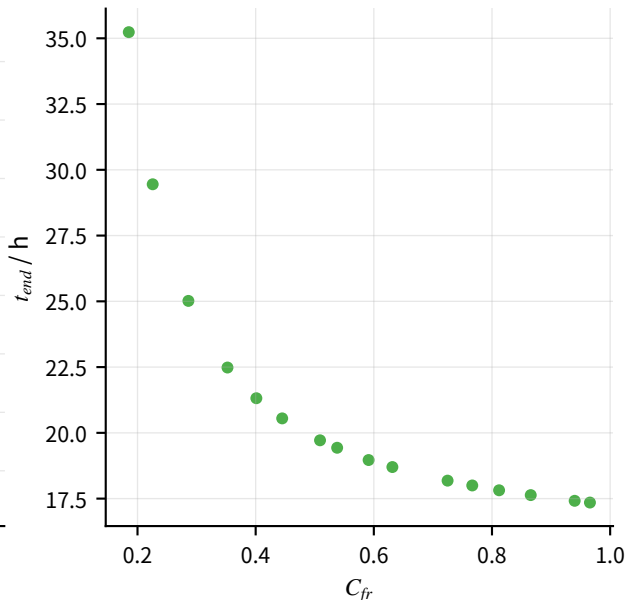
Sobol S1 vs ST



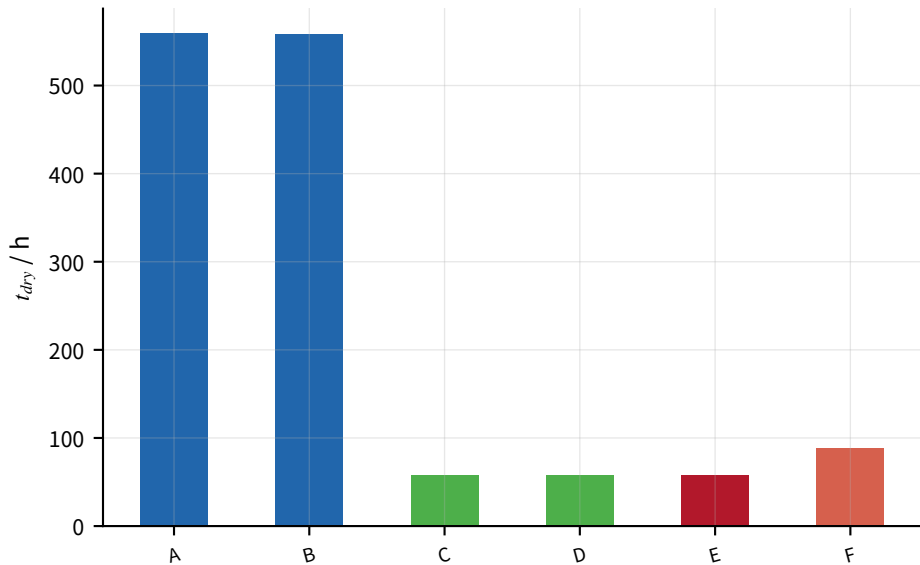
(a) 截断族曲线



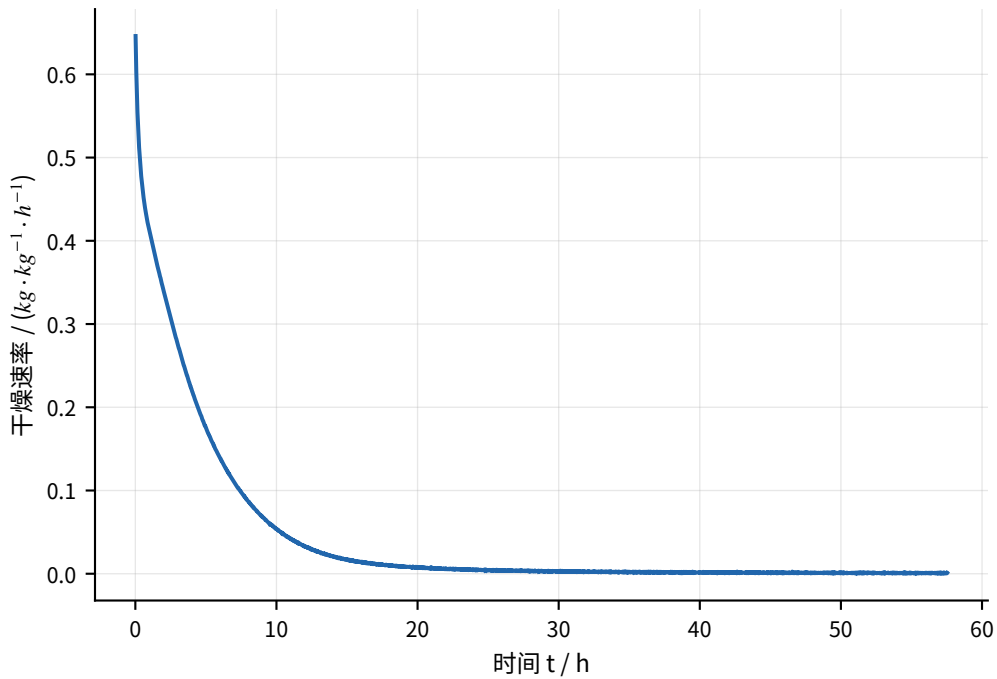
(b) LHS (n=16)



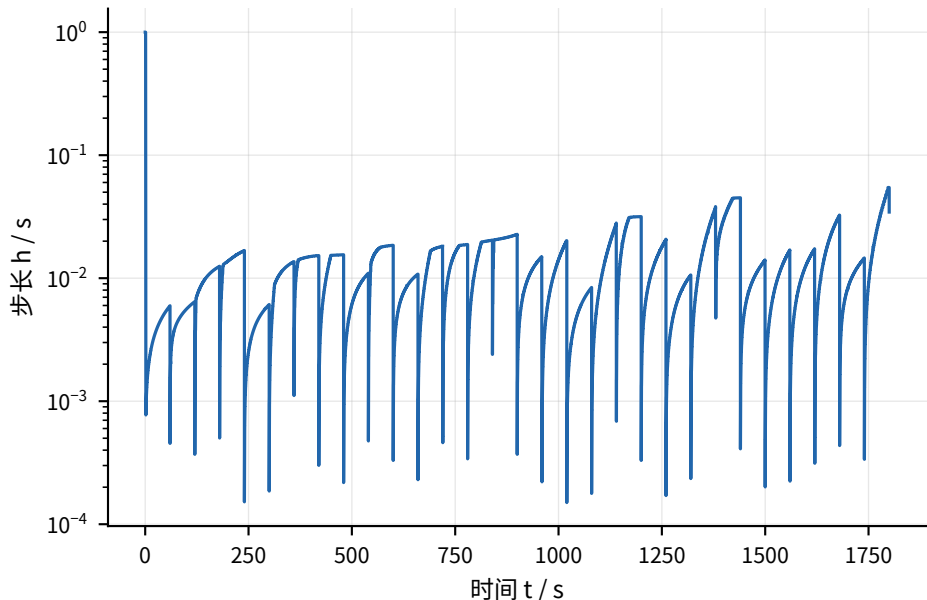
Q3 界面取法×网格分辨率对照



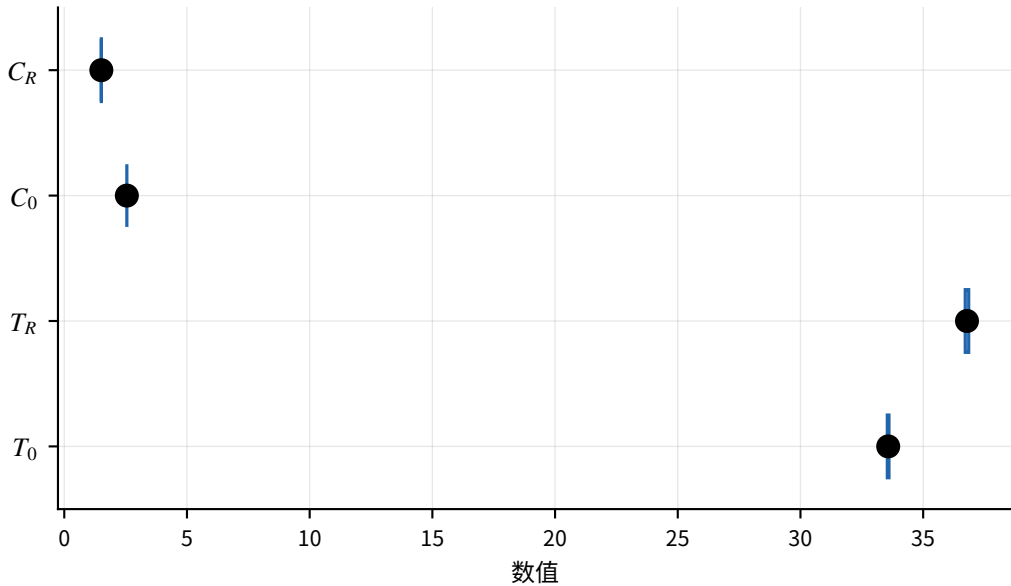
Q3 干燥速率曲线



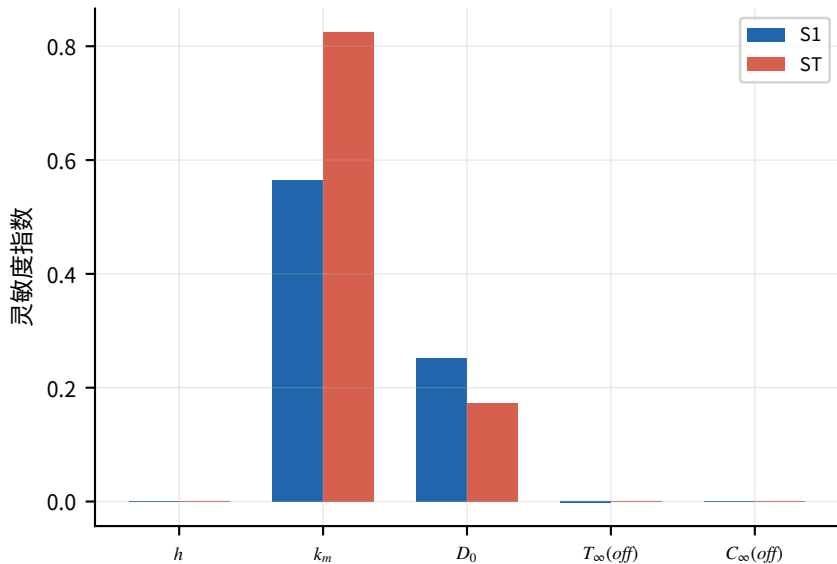
Q1 误差驱动自适应步长轨迹



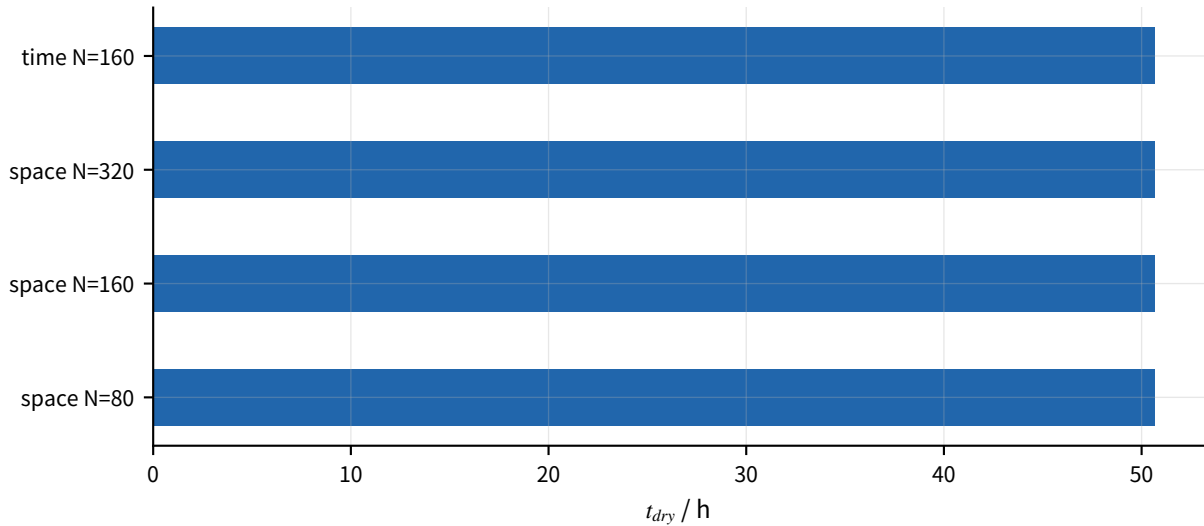
Q1 边界不确定性传播区间



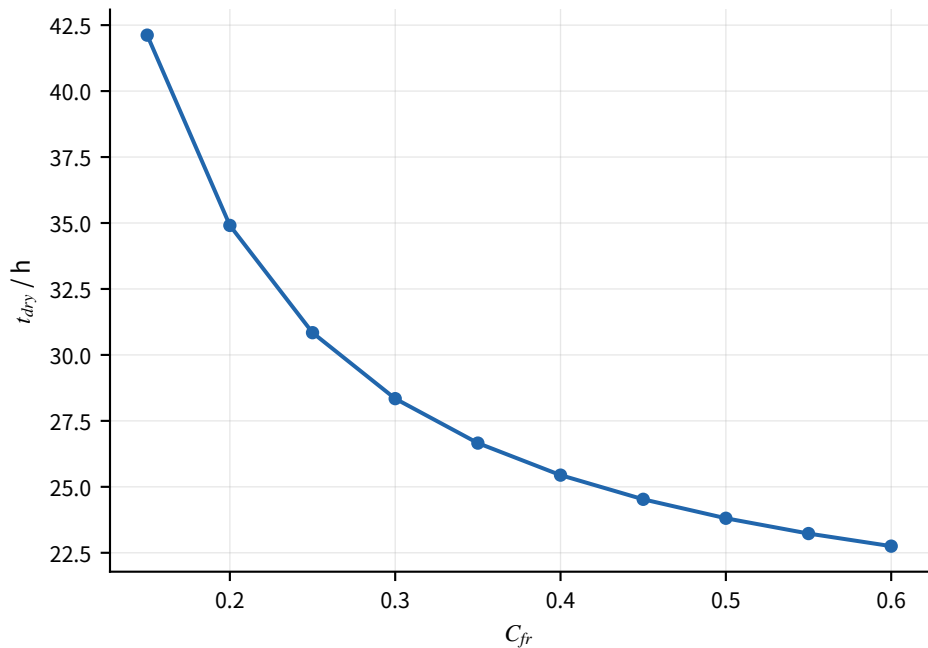
Q1 全局灵敏度 Sobol (CR)



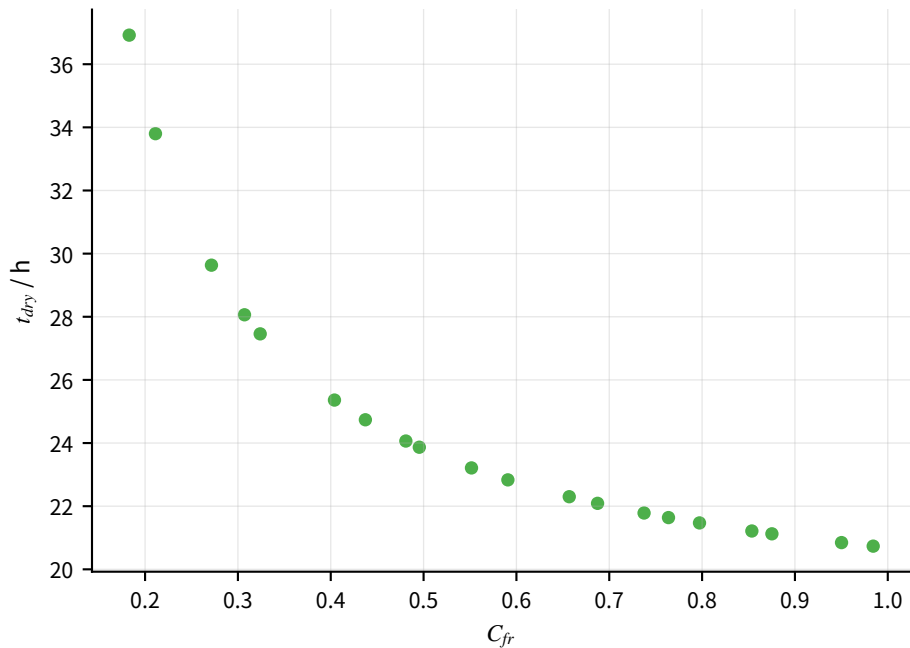
Q4 离散误差带 (GCI)



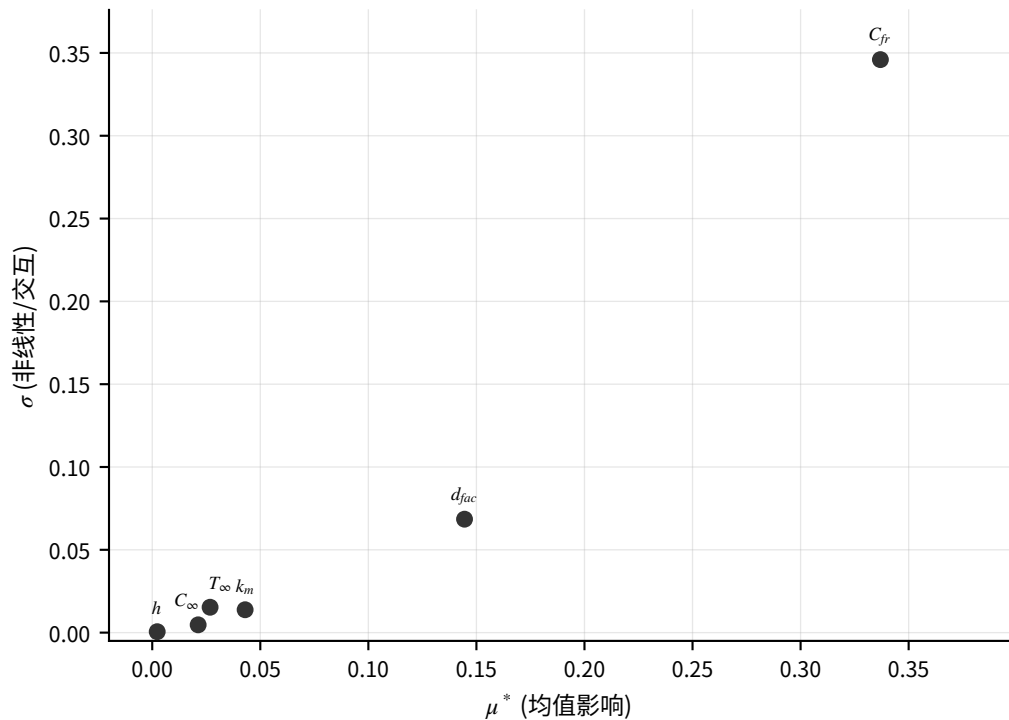
Q4 低C端外推不确定性 — 截断族曲线



Q4 低C端外推不确定度 — LHS分位 (n=20)



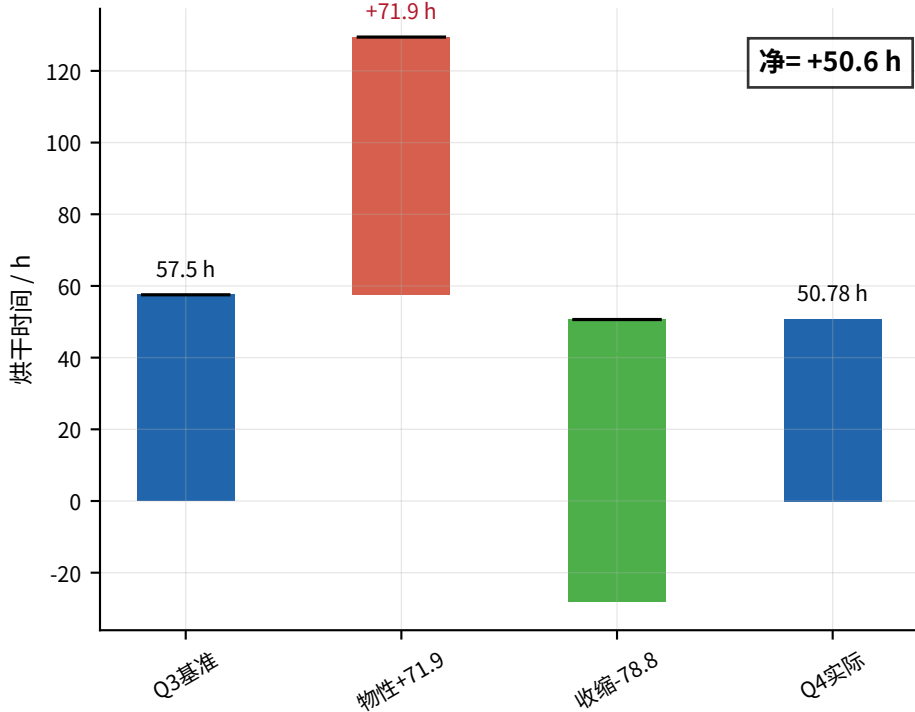
Q4 全局灵敏度 Morris (6参数)



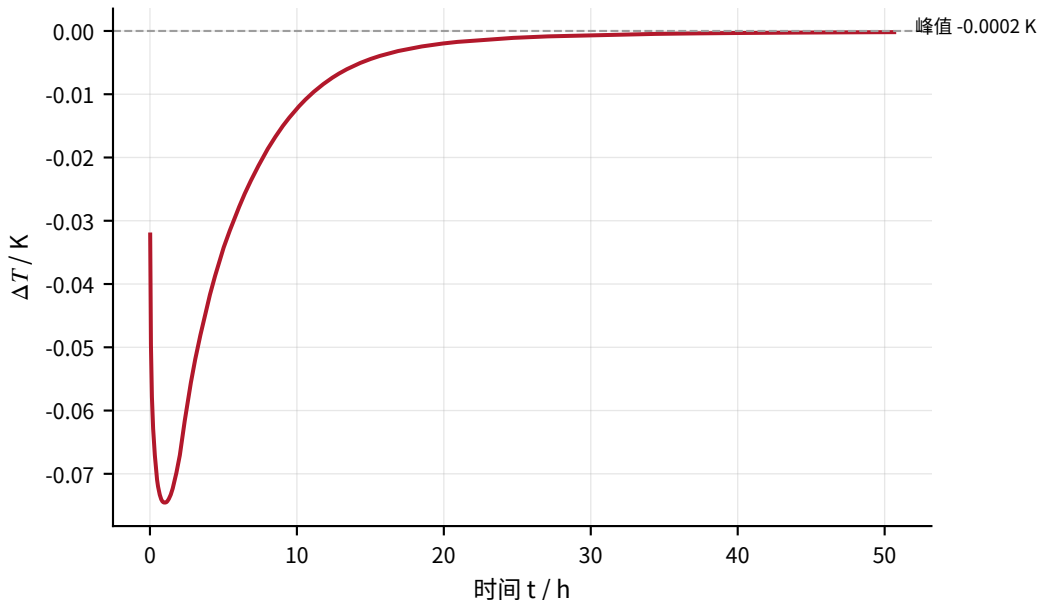
Q4 收缩-物性双效应分解

+71.9 h

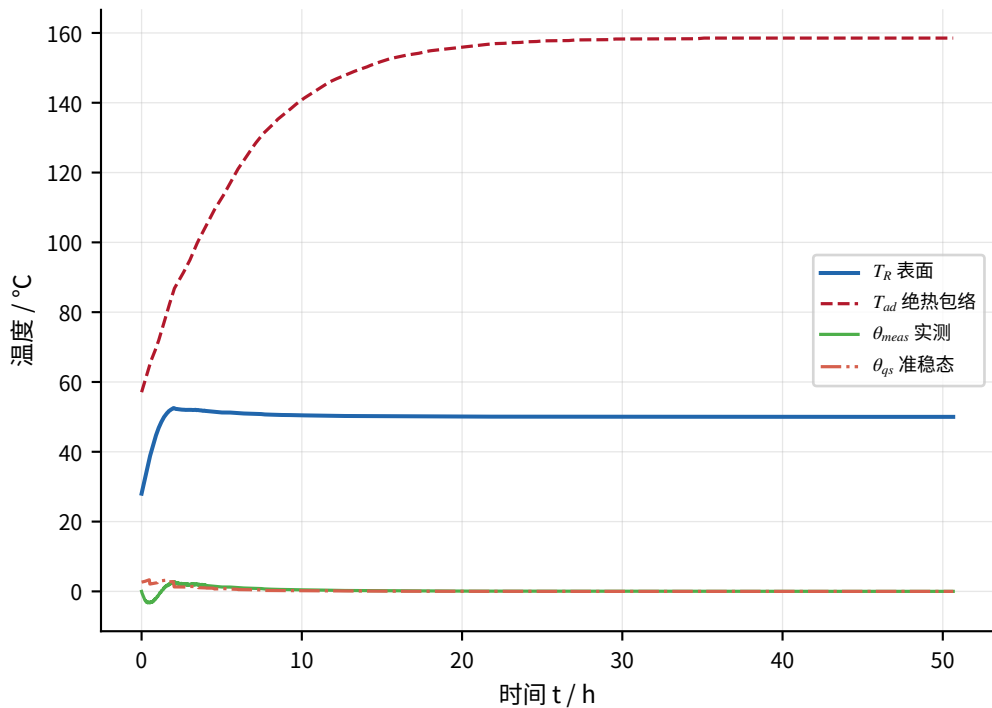
净= +50.6 h



Q4 潜热中段温降曲线



Q4 压缩温升机理与绝热包络



Q4 界面取法 × 网格对照

